

Hedging as a Liquidity Risk Management Strategy for Optimal Liquidation

No Diving in the Shallow End of the Pool

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Executive Summary

This paper examines the use of hedging to neutralise risk factor sensitivities as a liquidity risk management strategy during a controlled liquidation of an illiquid portfolio. We define liquidity risk in terms of the liquidation choice between an immediate negative price impact of being an outsized seller in a market, compared to the market uncertainty of a longer liquidation horizon running the Value at Risk of the market moving away from its current position. Hedging neutralises the market uncertainty but simultaneously introduces two further uncertainties: the basis risk and implied cost of carry for the hedged market risk factor and the risk of a funding mismatch in the mark-to-market of the hedge versus the underlying illiquid holdings. If the expected return on the underlying portfolio is higher than the hedged market risk factor cost of carry, and if funding can collateralise against the underlying holding, then it is an appealing choice to the investor. We look at how this can be extended to private assets in incomplete markets which introduces a further basis risk to the portfolio but still mitigates the market uncertainty in the case of private credit and fixed income-like private assets.

We start with a critique of the liquidity risk literature, which misses three critical aspects of the current state of finance which allows market-makers to operate in the first place: liquid hedges, funding lines of credit against the portfolio as collateral, and market-neutral portfolio construction resulting in the netting of risk from diversifying and offsetting hedging positions. Unless we are dealing with a default scenario and therefore a forced sale, all three of these should be explored in order to allow a reluctant seller to maintain the investment portfolio in place to ride out a temporary storm. A good example is the UK Government Bond storm following the Trussonomics debacle in October 2022¹, when fully funded UK pension funds were forced into selling liquid assets to fund the losses on their leveraged Gilt holdings that were neutralising their illiquid ultra-long-term pension liabilities. The focus of this paper will be how to solve these issues by looking at how to combine and net positions together, where one is illiquid and one is liquid.

We start by looking at the optimal liquidation of an underlying position and suggest some improvements in understanding the dynamics of the liquidation. This gives us a value at risk of the overall exposure, from a time to liquidation horizon mixed with a price impact of crossing the spread initially on excess trades that do not contain within the market volume. We then observe the risk-neutral trajectory that a hedge would allow us to complete and compare that to the accelerated trajectory of a less patient hedge, as the Value at Risk differential. We can then apply this risk budget towards a hedge as a cost of carry. We analyse the constraints imposed by differential liquidation trajectories from asset classes in the portfolio that have investment mandate target constraints, whereby the liquidation proceeds at the slowest rate once the mandate restriction is reached. With liquidation as a starting point, we can understand the liquidation constraint in the market volume, the asset mix constraints, the time horizon that we require, and finally the adjustments in time horizon that a hedge will enable.

We look at some of the maths behind the hedge, and ascertain that market uncertainty is a factor exposure, either rates, or equities, or currencies, and that a highly liquid, correlated instrument that represents the factor exposure can be taken as a negative exposure in order to neutralise the overall position; this facility also decreases transaction costs for market-makers, who will avail themselves of the same technique. When the underlying asset is illiquid, we look at how to equalise the timelines and look at the correlation boundary conditions that result from this extension of the timeline. We start with examining an exchange-traded instrument versus an over-the-counter instrument, to show how extending the liquidation framework to the longer instrument will result in a benefit. We then generalise this to more illiquid assets, and less complete markets, such as private credit and real estate, where factor exposures to rates, inflation, credit, and other macroeconomic variables help us to apply a counterbalance with appropriate hedge instruments. We also look at conditions where ownership of a private asset enables us to push down the hedge into the actual balance sheet of the asset, for example for a currency or interest rate hedge on the cash flows or liabilities level, attached to the actual corporate entity. We discuss the limitations of hedged liquidations as the constraints introduced by leverage and incomplete markets.

As a final deliverable summary of the techniques developed, we look at a Liquidity Report for hypothetical Global Rates fund facing a liquidation, and we compare the Waterfall approach of exponential unwind with the Hedged Hibernation approach of a linear proportional unwind. We ascertain that the risk reduction of the Hedged approach is about 80% for common market risk factors related to rates and credit in semi-liquid corporate bonds and structured credit.

We end this research with a discussion on the prudential ramifications of enabling a hedged hibernation of assets for liquidation, in order to reduce risk, as well as the market-maker benefits of introducing more active market participants from banks to act as prime brokers and clearing brokers in assisting with hedged liquidations. From a market perspective, we would expect the presence of such trading activities to introduce a new type of liquidity in the markets as well as bring down the transaction costs to the level of basis risk as a residual risk.

Motivation and Background

Regulatory risk frameworks have developed capital requirements to highlight liquidity risk management in Banks, Exchanges, Insurance Companies, Fund Management, and even Private Asset investors. Each have their own specific exigencies, but all require an action plan and liquidity management tools and analytics for managing a liquidation. The seminal literature that has evolved to meet these requirements has hitherto been focused on transactional mechanisms related to a single position, and optimally timing the exit from a position that is larger than the market will hold, as fast as possible.

Our main critique regarding the current academic literature is how far removed it is from actual market praxis being conducted by market participants, where hedging and funding lines are prevalent. Furthermore, a lot of the academic researchers hold consulting positions with market-neutral market-makers and statistical arb funds with market-neutral portfolios, which makes the contribution even more tenuous. Our view is that the current liquidation literature ignores three important aspects of modern portfolio management and market-making: the ability to hedge a portfolio with offsetting counter-positions in futures, options, and swaps markets; the ability to self-finance a hedging portfolio by collateralising the liquidation portfolio against the hedge; and the fact that there are already netting and diversifying positions in a portfolio that would complicate unwinding positions one by one. We look to introduce each of these with attendant techniques in order to enrich our understanding of liquidity risk management when faced with a timely and controlled liquidation.

Definitions and Delimitations

We define liquidity risk as the risk of implementation shortfall in liquidation which occurs when a position is too large for a single opposing market participant to accept a single transaction, thereby resulting in either a price impact away from the current market price, or a number of recurring transactions over several interactions during which time the market moves away from the initial price. This was first elaborated by Perold in his work on implementation shortfall and market slippage², and shows that it goes both ways, both in buying and selling securities. We further define price impact as the market microstructure of crossing and influencing the bid-ask spread, following the work of Diebold et al³. Finally, we follow the work of Almgren and Chriss⁴ where we view order flow as a random variable, so that we are dealing with two random variables, the market price itself, and the actual order flow that constitutes trading volume.

We define a hedge as an instrument that upon acquisition reduces the overall risk factor exposure of the portfolio, leading to a “market-neutral” portfolio that has less market uncertainty with an offsetting hedge addition. This will involve understanding the risk sensitivity of the portfolio vis-à-vis the hedge instruments: we reference the research to the Khan Interest Rate Calculus^{5,6} for hedging sensitivities in interest rates, as well as the Khan Optimal Currency Hedge Ratio⁷ for the mathematical approach to establishing a currency hedge. For long-only and market-neutral equities and the residual market beta, we refer to the foundational text by Grinold and Kahn⁸ on Active Portfolio Management. We look at some ways to collateralise a self-financing hedge with a clearinghouse, although mainly to elucidate the mathematics of extending the liquidation horizon. Not all exchanges do use risk margining, so they can learn from these methods to improve their capital efficiency and risk management as well. Most margin methodologies assume a clearing broker, who will manage funding, mark-to-market, and collateral. This is particularly important for private assets, where we endeavour to push down some of the hedges into the corporate entity.

Our language is mainly about selling but equally applies to buying. We delimit our study to a timely liquidation with no information asymmetries- there is a voluntary aspect to the timing and choice of the liquidation. A mark-to-market failure would lead to default for the investor, and hence an instantaneous forced liquidation referred to as a “fire sale”, with the horrendous price impact that would entail. Even in such a case, however, we could turn the attention to the market-maker, clearinghouse, or counterparty, that would effectively be faced with the same choices of hedging as this liquidity risk management strategy. An example use case would be a redemption at a Fund, whereby the investment manager would continue as a going concern, and would simply liquidate part of the holdings, whilst maintaining the overall investment portfolio and investment objectives in line with the agreed mandate for the remaining investors.

Another delimitation is that we do not look at algorithmic trading for optimal execution of trades in an order book to minimise short-term transaction costs. We are assuming stealth trading which will allow drip-feeding the trades with resilience so as not to create a permanent price impact, allowing the order book to recover and ignore any outsized trades that incur trading costs. For algorithmic trading, we refer the reader to Almgren⁹ as a separate field of study.

Market Microstructure Metrics

We commence by reviewing the timeline and mechanics of a single security holding being sold off, with some tweaks to the various models found in the market microstructure literature. First, we determine liquidity constraints from the position size versus the market size for the security. For this we look at some liquidity measures, which will give an idea of the market absorption rate and the time to liquidation and hence the time under market uncertainty where the market can move away to a more adverse price; and transaction measures, which will give an idea of price impact from taking more liquidity out of the market by paying a higher price. Finally, we look at parameterising the market uncertainty in a measure of dispersion and see how a hedge reduces this dispersion the more correlated the hedging instrument is.

Liquidity Measures: Market Size and Trading Volume

Market Size Float: Total Openly Available Stock of an Asset

The first general concept of liquidity is Market Size: how many securities available for trading or overall market value that is being traded on a market, termed the Float. For equities, this measure is all the stocks that have been issued that have no restriction on trading. For bonds, this measure is all the bonds that have been issued. For exchange-traded instruments, it is every contract that is generically similar across different terms (maturities) that is being traded. For FX, it is all OTC FX Forwards that are similar across different terms, that are not netted out, that are being held. For OTC derivatives, it is all non-netted contracts of a particular underlying security that are available. Note that for OTC derivatives, the concept of non-netted contracts is important, because the netted contracts are neutralised and carry no risk if cleared, basically hedged against each other. They can be treated cancelled contracts; this is the basic idea of hedging for liquidation that we wish to achieve for other positions in the course of this study. Finally, for private assets, take Real Estate, the float would be the housing stock available for rental or sales, so not the owner-occupied assets that are not available for sale. This concept separates the float from usage; thus, the asset is either an investment asset or being exclusively used and not for sale.

Open Interest

Open Interest is a term reserved for exchange-traded futures and options, which refers to the open net exposure in a group of instruments across different terms. Exchanges restrict the amount of Open Interest that a single market participant can hold, which makes it a particularly good measure of Float for an Instrument, because it is the likely market you can attain. It is also a good measure for a cap of a position, say at 5% of Open Interest. So, the measure plays into the actual design of the position: it says a maximum achievable position beyond which there would be a constraint on liquidation from lack of counterparties.

Trading Volume: Traded Asset Amount Over Time

The second related concept is Trading Volume: the actual Float that is traded over a given interval, usually daily. This is because the Float does not give us a clear measure of what is actually available to trade. A highly valued company with a small float can be illiquid when measured in terms of Trading Volume. A corporate bond likewise might have been issued but then ring-fenced against future liabilities and held separately in an insurance company's balance sheet, and then held until maturity.

For this reason, Trading Volume is the correct measure for the liquidity of an investible asset, and what we will use when considering liquidation. For exchange-traded instruments like equities or futures, we have a clear daily trading volume available as a statistic of the exchange. For OTC instruments like bonds, FX, or any derivatives, we need to infer the actual volume, by estimating the figure for clearinghouse statistics, or regulatory statistics. Also bear in mind, for OTC we need to have multiple counterparty trading lines to access the trading volume, so in practice, it is sensible to limit the measure for Market Volume to that available from authorised counterparties.

Another note is that trading volume should be measured at the stable volume that is assessed during normal market times in the recent past, as an average measure. In stressed markets, trading volume does indeed increase by a multiple, but so do trading costs and liquidity spreads, so they are not accurate measures for completion of trades. Trading volume is a rate variable, and traders refer to the change in rate to identify “slow markets” or “fast markets”. This will refer to the amount of information that is being impounded in the price, often resulting in large changes in price; for the purposes of liquidation, we are interested in the average stable market volume as a proxy for the time taken to liquidate a position.

A common trader's heuristic is that we can take 10% of the Trading Volume “without leaving a footprint”- without having an endogenous price impact from the trade. Similarly, the common trader's heuristic for the maximum absorption of the Trading Volume with Maximal Price Impact taking the whole Order Book in a position “dump” is 25% of the Trading Volume, before the other market participants start to front-run the trade and thus creating “chock-a-block”, a dysfunctional market. In an exchange-traded market, these limits are often hard-coded into the Open Interest limits to prevent positions that create this effect. We will call the share $k\%$ that we trade out of the Trading Volume our “Cut” of the Market, and we will call the rate of liquidation for the unwind of the position our “Chase” rate; the chase rate may vary over time. Hence, for a chase without footprint, we would take a Cut of $k=10\%$ of the Trading Volume, equal to a Passive Chase. Divided by the Position Size w_{t_0} , that gives us a “Liquidation Horizon” and “Unwind” rate.

$$\begin{aligned}\kappa \text{TradingVolume} &= \text{Chase}_\tau \\ \frac{\text{Chase}_\tau}{\text{PositionSize } w_{t_0}} &= \text{LiquidationHorizon } T \\ \text{Unwind}_\tau &= \frac{1}{T}\end{aligned}$$

Transaction Measures: Bid-Ask Spread and Order Book

Bid-Ask Spread

While the theoretical price of a security is the Mid Price, it does not exist: the price is quoted either as the Bid at which an investor is willing to Buy the security or the Ask/Offer at which an investor is offering to Sell the security. On average, it will be the Mid Price, as buyers or sellers cross the spread in order to take liquidity, thus we take half of the spread as the transaction cost. In private markets or auction markets which often require a broker intermediary to connect buyers and sellers who then negotiate for the price, then the sales commission can be treated as effectively being a bid-ask spread.

Half the Bid-Ask Spread as a Price Effect is the Transaction Cost

The quoted spread is not necessarily the transaction cost: it is the final effect on the price which is the transaction cost. For example, let's say the Bid-Ask Spread on a 10Y Swap is 1 bp at 0.01%, so the quoted transaction cost is 0.005%, then the actual transaction cost is 0.04%. For any bond or swap where the Bid Ask is quoted in terms of the yield, we can use the Khan Par Yield Duration formula as a quick and easy measure to calculate the actual transaction cost from the spread quote in percent as a negative number to be applied to the weighted return:

$$TransactionCost_{\%} = Duration \frac{Ask_{\%} - Bid_{\%}}{2}$$

Transaction Costs Are Lower with Higher Trading Volumes

The more trading volume we have in a particular asset, then the higher the liquidity and hence the lower the transaction cost: the narrower the spread will be. Spreads are dynamic: in stressed markets, where there is more market uncertainty, the spreads will widen, while in slower markets where there is no new market information, the spreads will tighten. The actual transaction cost can be anything from 0.0001% in a high-volume market like FX, to 10% in a unique private asset, like a private asset transaction. Again, market volume will most certainly be a decisive factor for the transaction cost applied, as a higher market volume offers the opportunity to enter into a trade at a later date with minimal market uncertainty. We will look at how to establish a boundary condition between transaction costs and market uncertainty in the next section and later how offsetting correlated hedges reduce the spread.

We only need to cross the Spread and thereby incur transaction costs for amounts beyond the Cut of the market that we can enter by passively waiting for prices at our Ask. Up to the Cut of the trading volume, we can have a passive Chase. For an amount beyond this, we need to cross the Spread and take the Bid, hence incurring a transaction cost. We will refer to the share of price taking orders as the active Chase. The order flows in terms of Bids and Asks arrive independently in continuous time, creating another random variable that we will look at when creating the trading schedule.

Transaction Costs Is Market Uncertainty by Square Root of Time Process

Illiquidity leads to longer trading horizons and hence longer exposure to market uncertainty, which grows by a square root process. Hence the larger the transaction size, the larger the Bid Ask Spread. Each instrument will offer a different bid-ask spread for different transaction sizes, hence forming a transaction cost vector. For example, we might have a bid-ask spread for different trades, ranging from 50M to 500M. Any amount beyond that, would involve negotiation to understand any information asymmetries implied in the trade. Most empirical research, such as Torre and Ferrari in the continuously updating Barra Market Impact Model¹⁰, has found that the Transaction Cost as implied by the Bid-Ask Spread is a power function with relation to the Order Size, such that $TC \approx \beta \text{ OrderSize}^\gamma$, where γ is found to be around 0.5. We will later propose that the square-root process of Transaction Cost to Order Size relates to the Market Uncertainty from holding a position over an ever longer holding period that increases the Value at Risk from the resulting liquidation period to $\sqrt{\tau}$. The larger Order Size results in a longer prospective Liquidation Horizon, and the Value at Risk of the longer Liquidation Horizon increases by $\sqrt{\tau}$. Equally, any attempt to demand a larger transaction cost, will incentivise the seller to hold on to the position and face market uncertainty as opposed to the transaction cost. Hence there is a boundary condition due to the choice enforcing economic alternatives. In the next section, we consider this as a “Good Deal Bound” which brings a parity between the choices, firstly against market uncertainty but even in more complete markets against hedged instruments with a resulting narrow basis hedge risk governing the transaction cost, which also increases by $\sqrt{\tau}$.

Order Book: Market Depth

While the Bid and Ask are where the price is found for a buyer and seller, the prospective buyers and sellers are found in ever increasing numbers the further from the market price we find ourselves. In exchange-traded markets, this is fully transparent from the Order Book. Here, we can see the ever-increasing Buy and Sell positions further from the actual Bid and Ask. Traders refer to this as “market depth”: the more bids and offers for a given instrument, the deeper the market is

We can get an idea for the overall price impact for taking liquidity, from how many layers of bids we have to go through in order to liquidate an entire position. Typically they are shaped as two Gaussians, one on the Bid and one on the Ask, with small orders closest to the last price, then larger limits on either side. To take liquidity and hence incur a transaction cost, a seller would put in a Market Order, and fill the order at the successive bids until the order is filled. To avoid a transaction cost, a seller would drip-feed a position as the lowest Ask being offered in the order book, and constantly replenish that position. To create a positive return, and hence a transaction profit, a seller would place a Limit Order that is above the lowest Ask, for the order fill, and expect a Market Order to take the first ask as well as the limit placement. Market Orders arrive independently and randomly to the order book, so for positions that are within the right size, it is a feasible way to fulfil a transaction. We will see in setting the trading schedule that taking advantage of this phenomenon of order arrivals enables an aggressive approach to market-making, such that it is advantageous to always be present on both sides of the order book both taking and making the market.

With an inordinately large position compared to trading volumes in illiquid assets, the order book will not cover the position, or at such an exorbitant transaction cost as to make it unreasonable. A common practitioner approach for hedging is to impose a “Good Deal Bound”¹¹ as documented by Cochrane and Saa-Requejo. This is normally done in incomplete markets to determine a relative value position in a hedge against an underlying, but we intend to do this for transaction costs. In terms of the market microstructure, that means filtering for reasonable bids and asks in the order book and otherwise waiting. In terms of economics, it is a way of transferring the profit function of an investor to a market-maker and vice versa. By understanding the transaction cost and the market uncertainty in the same equation, we can also say that it applies to the buyer as well as the seller, except given the asymmetry in position size for the resulting position.

A simple way to implement a Good Deal Bound is to weigh the transaction cost against the market uncertainty, as measured by the Value at Risk, at a level suitable to an investor’s risk aversion, at which the investor would be indifferent to waiting over time. Given a liquidation in a stressed market, we can assume a period of high kurtosis, so we will avail ourselves of the Khan Kurtosis-Adjusted Value at Risk KVaR¹² as a “Good Deal Bound”. A suitable heuristic for the VaR confidence level would be 95%, or the worst day in a month. For longer periods with illiquid assets, no volatility estimator scales well from daily measurement, so we will normalise it further with an exponential decay towards a Normal Distribution. This is well documented in Time Series Analysis literature on empirical finance, see for example Diebold et al¹³. Thus, we now have a way to compare the Transaction Cost to the Market Uncertainty, which we will term the Khan Good Deal Bound for Transaction Costs versus Market Uncertainty:

Khan Good Deal Heuristic for Transaction Costs versus Market Uncertainty

$$TransactionCost_{\%} \sim KVaR(1-p) = \Phi(X < p) = -Z_p \sigma_K, \text{ where } \sigma_K = \sqrt[4]{\frac{K_{**}}{3}} \sigma_*$$

Where we set $p=32\%$, indicating a left tail outcome that is a bad day, or $Z_p=1$, as a suitable heuristic level of risk. The lower the market uncertainty, the less likely we are to feel pressure to trade, and equally the lower the transaction costs will be. We do not call this a Bound, but rather a Heuristic Approximation, creating a shape in the economic choice for the position holder.

For a market-maker, or a hedged position, we would assume that the volatility will not be of the underlying position, but rather of a hedged position, that combines both the underlying position and an offsetting hedge. We will look at this in the next section, and as a final economic theorem. In the aftermath of the determination that hedged positions reduce the market uncertainty factor, it would be expected that transaction costs could be effectively reduced to the level of the residual basis risk of the outright positions offset by the hedge positions.

Market Uncertainty: Volatility

Standard Deviation of Daily Returns

As detailed in the earlier section, our measure for market uncertainty is the ex-ante measure of dispersion of returns over the period of liquidation. We can use standard deviation over the correct time period scaling expressed as an annualised volatility, or better yet, we can adjust the kurtosis for the returns, to apply a K-Vol to the returns distribution, which we'll express annually as well, but bearing in mind that we wish to measure it at the appropriate time scale. For illiquid assets with a time horizon of 6 months or longer, we can safely estimate a normal distribution for the returns, but equally that the estimator is much less efficient, as is documented in Time Series Analysis literature, see for example Christoffersen and Diebold¹⁴.

Khan Kurtosis Adjusted Volatility Estimator

For our purposes, we make use of the Khan Kurtosis Adjusted Volatility Estimator¹² as a measure of market uncertainty in daily returns:

$$\sigma_K = \sqrt{\frac{4}{3} \frac{\kappa_*}{3} \sigma_*}$$

And for scaling in longer-term forecasts for illiquid assets, we use larger time increments which normalise the returns, and we decay the kurtosis over a 3-month time period to achieve normalised kurtosis in forecasts. We keep the notation, with the understanding that $k=3$ for longer-term forecasts.

Hedge Basis Risk from Optimal Hedge Ratio

Another market risk term we should introduce at this time, is the residual market volatility after a hedge position is applied. Given incomplete markets, no hedge will be a perfect hedge, nor does it have to be. We will assume hedges that reduce the market volatility, and that they do so by having a strong correlation with the underlying asset returns. Hence a hedged portfolio will have a basis spread between its long underlying position w and its short hedge position $h=-w$, that results in a residual market volatility equal to the hedge basis risk:

$$\sigma_\varepsilon = \sqrt{w^2 \sigma_\pi^2 + h^2 \sigma_h^2 + 2w\sigma_\pi h \rho_{\pi,h}}$$

$$\text{Setting } h^* = -w_\pi \rho \frac{\sigma_\pi}{\sigma_h}$$

$$\sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

This can also be expressed as $h^* = -w_\pi \beta_h$ as a Hedge Factor Beta which neutralises the portfolio exposure against the hedge factor. Then the term $1 - \rho_{\pi,h}^2$ can be understood as the Explained Sum of Squares ESS in an OLS regression, and the σ_ε as the residual error term. We will review this more formally later in the hedging section.

Khan Good Deal Heuristic for Transaction Costs versus a Hedged Position

One for the Money, Two for the Show, Three to Get Ready, Four to Go:

$$TransactionCost_{\%} \sim 4 \cdot KVaR(1-p) = \Phi(X < p) = -Z_p \sigma_{\varepsilon}, \text{ where } \sigma_{\varepsilon} = w_{\pi} \sigma_{\pi} \sqrt{1 - \rho_{\pi,h}^2}.$$

(note: 4 times because of the additional round-trip Hedge Trade means three transactions, and the extra is a reasonable “Good Deal” margin for the time and effort)

The lower the Hedge Basis Risk, then the lower will be the Khan Good Deal Transaction Cost Hedge Heuristic, and hence the more likely we are to be patient and market-maker transaction costs are likely to be low. This indicates why highly correlated positions such as rates will have lower transaction costs. This relationship will serve as a foundational understanding for a underlying theorem both as regards transaction prices as well as the motivations for holding a hedged portfolio with a longer liquidation period. Note that the transaction cost will be 3 times the basis spread at the minimum, because the spread is created by a round-trip trade in a hedge that increases the number of transactions to 3. Setting the Margin at 4x, seems a sensible heuristic economic approximation to enable profit and incentivisation around establishing the position. This is also a reason why we do not consider dynamic hedging as a hedge; a dynamic hedge would involve a theoretically infinite number of transactions. A practical example is the “portfolio insurance” debacle in the late 1980s by some of the leading thinkers in academic finance, where slippage and costs led to systemic disaster. Finally, the 4x might be incorporated in that the p-value might be lower such that the Z score is higher in the case of a commercially-minded market-maker, maybe such that the hedge basis risk is not achieved more than once a month or as frequently as the competitive pressures would allow, so a Z score indicative of a VaR95% perhaps might be a good indication. The Kurtosis adjustment for an error term or a Basis Spread, is ALWAYS higher than for a single variable, so we can expect the Kurtosis adjustment to be higher for any estimate on Hedge Basis Spread.

We will revisit this as an Economic Theorem at the end, where once we’ve developed a hedging methodology for liquidation of asset managers or investors, we look at the incentives for a market-market or bank participant to get involved with the proper incentives set by this Khan Good Deal Heuristic determining the actionability for market-making in less liquid assets with the purpose of liquidation. Given a regulatory framework where asset managers, insurance companies, or non-financial institutions cannot do this themselves, and where further regulatory directives limit the profitability for banks in more simplistic activities, we can expect that the ability to transfer less liquid fixed income securities with appropriate hedging mechanisms becomes a new area of growth and profitability for banks. With that in mind, the Do-It-Yourself approach of hedging for liquidation, and liquidating via a Market Maker and thus incurring transaction costs, becomes the same in terms of cost: one implies the other, so we approach more of an equilibrium equation between the two as more people develop the methodology outlined in this work. The economic incentives of one, complete the other, so we approach complete markets and thus greater liquidity as a result. The gap will be narrowed by technology, as open banking standards, automated processing protocols of standardised data attributes in financial contracts, and fintech channelling for funding syndications and rates marketplaces will enable bank to have a wholesale market for this type of funding.

Liquidity across Asset Classes: Illiquidity is the Norm

Now that we have a taxonomy of the metrics that we will use for the measurement of liquidity, let us first investigate how liquidity comes to be an attribute in the first place. From a philosophical perspective, it is the exception, not the rule. Liquidity means the open float and the number of participants, and hence it means size. And size is a “bestseller effect”, following a power law. Every asset class has a spectrum of liquidity, and that spectrum means small to large. But we often forget that the small enterprises and households dominate the large portion of financial activities, and that it is the role of banks in funding and market-making that intermediates that liquidity, especially in lending activities. Any large owner makes an asset illiquid; the trick is to turn it on its head and make the illiquid small in comparison through multitude, which is what a banking book or insurance book does. There is a level of intermediation that disappears at the smaller level: an underwriting desk is required, an analytical expertise of the business. These are taken for granted in liquid securities, we all assume someone has done it before us.

Equity Concentration Follows a Bestseller Effect

Equity is a good place to start: there are only a few thousand companies out of millions in the global economy that have liquidity suitable for the largest investors. The top 10% of companies in a stock market index account for $\frac{2}{3}$ of the open float in the market capitalisation¹⁵. Since these are perpetual into a time of default or consolidation, we assume a liquidity that is often illusory in everything but the largest hundred floats globally. It is common in smaller economies to see a “2 weeks in, 2 months out” strategy in terms of buying and selling even for big floats. Beyond the largest capitalisations, the listing is often nominal and would not support active trading.

Cash Flows and Growth Prospects are King

The biggest problem with Equity, is that cash flows and growth prospects decide the price. The actual discount rate is hard to isolate as a decisive variable. This means that idiosyncratic risk is all that matters. It’s very hard to replace one stock with another, there is very little commonality between different stocks, making them highly uncorrelated as businesses and investments. We’ll see the opposite effect in Fixed Income subsequently.

Index Bundling Helps Equity Liquidity

Passive index tracking by global asset management firms, “packaging” equities in an index “bundle” has resulted in spreading liquidity across more issues¹⁶. Thus, the idea of aggregation and bundling of securities, is a key to understanding how liquidity can work for smaller securities. Furthermore, an index bundle helps to signal an uninformed, passive trade, hence reducing information asymmetries that might affect pricing. It also serves as a natural diversifier, since the trade will then effectively have a correlation of one against an Equity Index Future, enabling a market-maker to do a hedge a market-neutral block trade against a liquidation. If this dynamic applies to Equity, then surely it will apply to more granular securities which by their nature are not scaleable to the grand scale of stock market issues- with this in mind, let’s look at Fixed Income.

Fixed Income: 50 Shades of Disarray

Fixed Income is essentially illiquid because of its buy-to-hold nature: even the most liquid government bond auctions have been manipulated and rigged systematically¹⁷ and those are the most liquid “On the Run” Sovereign Bonds¹⁸. Fixed Income is also OTC via dealer banks as opposed to on an exchange, so they lack the central price discovery market-mechanism of an exchange and clearinghouse, making pricing and data history serious issues. The debt of the Firms with the largest Equity issues by Market Capitalisation, will obviously also have some related debt issuances, but by construction even these are restrictive due to the granularity imposed by maturities and covenants in the terms and conditions of the obligation, so they need to be acquired on the “When Issued” market, hot off the press, after which they are often held to maturity by asset managers with long-term liability matching objectives. Even when traded, only $\frac{1}{6}$ ever changes ownership in any given week, and it takes a full year for any given issue to change owner¹⁹. Most new issues are “refinancing” bonds, so that the liability side of the balance sheet keeps rolling over usually at a 5 year term, and corporate bonds are callable to refinance at lower levels of rates should they arise. Government bonds have a market only when they suit certain attributes, which explains the yield curve having a negative “convenience” yield in the far end, in that longer duration matches well against long-term liabilities of pension funds and insurance companies, and are then ring-fenced and held against the liabilities such that there is no market for them. The same for corporate bonds, which are invariably held to maturity. 80% of corporate bonds that are actively traded are financial bank bonds, that are essentially securities on the banking book.

Bond Classification Hierarchy

Bonds have a clear classification hierarchy from most liquid and highest credit quality, but the lowest are not necessarily lower credit quality just lower liquidity, giving a grouping by Type of Issuer. In terms of regulatory measurement, they are called Credit Quality Steps.

Government Sovereign Bonds

These are the highest credit quality, since they can be issued and paid by sovereign issuance, and the most liquid since sovereign bonds can be used for collateral against cash. Other bonds in this grouping are those affiliated with the sovereign issuance, for example local authority, municipal, or super-sovereign bonds. These types of bonds still constitute 60% of the outstanding bond market (ICMA statistics). From a liquidity perspective, they are treated as the highest liquidity, given their ability to be collateralised, so from a regulatory perspective, they are treated as cash (High Quality Liquid Assets²⁰).

Corporate Investment Grade

At a similar level, are corporate investment grade bonds with senior liens on the company assets, accounting for 25% of the market: 80% are financial bank debt issued by banks against their banking books. This is important to recall when we look at other types of private loans and securitised products, which are effectively banking books as well. Other mainstay corporate investment grade relate to the major corporate capital structures discussed in Equity, mostly in Industrial, but lately across all Sectors, these will have an Industry or Service Sector Classification that is important in aggregating risk across risk factors. Finally in this section we find Utility bonds, which are secured against a relatively stable income stream from the utility revenues, which can be regarded as a quasi-sovereign asset.

Corporate High Yield

Corporate high yield securities are either “fallen angels” that are originally issued as investment grade bonds which have fallen in terms of credit quality, or they are Leveraged Buy-Out speculative debt, whereby funding is raised for an acquisition of a company in view of asset stripping the takeover target by issuing dividends to pay back the leveraged nature of the acquisition, such as what happened to Thames Water utility company in the UK. Generally, this is private equity type of financing, but sometimes banks are involved as well, such as the Macquarie takeover of Thames Water.

Securitised Credit / Structured Credit

Structured Credit means pooling thousands of small private loans in a Special Purpose Fund/Vehicle which then issues securities which are listed by a net asset value and can be traded. This is the least liquid and smallest area of Fixed Income at about 10% but most interesting area to discover, where we can expect to see a lot of growth and transformation over the years. In many senses, this research paper empowers us to look at this segment differently. We expect that fintech intermediaries, open banking protocols, automated straight-through back-end processing, and better standardised data structures with improved look-through will revolutionise the ability of structured credit, to the point of supplanting or more greatly supplementing the Financial Bank Debt seen in Corporate Investment Grade. There is very little difference between them, other than liquidity. Let’s look at the various sub-divisions.

MBS Mortgage-Backed Securities Passthroughs

Mortgage-backed securities are pooled retail house loans from various bank issuers that are passed through a quasi-government agency that packages them into securities against the collateral value of the properties. This is typical for America, and probably a good model for other countries as well. The typical model for American MBS is that they are Fixed Rate 30 years callable bonds, such that the obligor can call and refinance the loan at lower rates should the market move lower against them. This is by far the biggest area, about 20% of the market.

CMBS Commercial Mortgage-Backed Securities

Commercial MBS are non-agency backed (some are recently agency backed) but have a larger corporate entity such as a real estate developer, sometimes Equity listed, who is managing the loans for development of a housing project, involving office complexes or large residential development projects. These tend to be very well collateralised and over the course of a development, but with some more sector concentration or geographical concentration, while MBS tend to be more diversified and with more obligors.

Covered Bonds

Covered Bonds are again property collateralised loans like UK debentures or German (and recently French) Pfandbriefe securities that are similar to CMBS, with some regional differences, mainly related to variable rate as opposed to fixed rate in the base rate.

CLOs Collateralised Loan Obligations (Leveraged Loans)

CLO are the structured credit equivalent of High Yield Bonds, generally speculative grade commercial loans against companies taking more debt against their assets but at a senior secured lien to other loans. These are often tranching into risk levels, with a default waterfall in the SPV, whereby the highest grade CLOs are paid first on interest payments, and the final equity tranche receives any residual payments that remains. CLOs tend to be overcollateralised to prevent excess default. But they are speculative but senior secured loans to companies.

ABS Asset-Backed Securities (Household Finance)

ABS are retail household finance pooled loans covering auto, credit card, student, second house loan debt, mainly to finance various stages of the household lifecycle, against second lien collateral, for example consumer durables like a car. With the emergence of pooled car ownership and self-driving autos, we can again expect this segment to take on a whole new importance in the household economy.

CDO Collateralised Debt Obligations

One relic of the Great Financial Crisis of 2008 was the CDO²¹. These were like CLOs, with tranching to create synthetic Credit Quality buffers in a Waterfall payment structure, but with the actual loans having junior subordinated liens against various retail loans through several layers of tranching (tranches on mezzanine tranches). There was furthermore a greater speculative credit grade type of obligor, which were called “sub-prime”, in other words obligors with a poor credit history and poor prospects for interest rate coverage of loans. The ones that caused particular trouble in the 2008 crisis were CDOs on CDOs, so they were aggregations of the mezzanine tranches of other CDOs, such that being over-collateralised, they were supposed to act like a senior lien (which they did not). These have fallen out of favour, but it is important to have look-through in both CLOs and ABS, for characteristics of mezzanine tranches being passed off for something else or sub-prime credit quality in the loan origination. The main problem with the 2008 crisis, was that all the loans defaulted with no waterfall to be seen, from lowest to highest grade, like the Victoria waterfall in Zambia drying up during a hot summer. So, the so-called AAA investment grade tranche was as worthless as the junk tranches: of the \$400B in write-downs of CDOs during 2008, \$325B were so-called AAA²². The basic learning from this, is that a tranche of a tranche retains the same credit quality: there is no diversification benefit of owning several mezzanine tranches- they don’t magically turn into AAA. The ratings agencies and the top banks at the time assumed they did, which is a statistical fallacy.

Classification: Be the Bank

A quick review of the bond classifications from most liquid to least liquid, shows that the most liquid Financial Bonds, and the least liquid Structured Credit, are actually the same. They are both originated from Private Credit, and then either structured as a loan by the Financial Intermediary, or against an SPV funded vehicle. This is important to note: they are both the same private credit issuance. The one thing that might separate them, is what is called the “pipeline problem”, namely that the asymmetric information implied in the Bank’s underwriting of the loan, means that they are inclined to retain the good obligors and spin out the sub-prime obligors, hence “keeping the good apples”. With great look-through and fully automated processing under an open banking protocol, this is becoming less likely. The premise of developing a strong liquidity methodology is how a fully funded investor together with a self-financing hedging mechanism can complement bank activity.

Credit Quality and Bond Ratings

Credit quality comes down to two principles: Collateral (Loan to Value) and Net Income to Interest Expenses (Net Interest Coverage). Lacking assigned collateral pledged to a bond, then seniority of the bond against other issues takes precedence in determining a waterfall of asset value being allocated to a bond by senior lien. And for Net Income the stability of the revenue and income stream decides the probability of always meeting payments. Ratings agencies generally analyse these two attributes in order to arrive at a probability of default, which in turn determines the credit spread applied to the valuation of the bond. In broad terms, we arrive at “Investment Grade” AAA-BBB which implied a Credit Spread of 35bps to 150bps depending on the lien, “High Yield” BB-CCC 300-500bps, and “Restructuring in Default” which can be anything. From an investment mandate, many managers are restricted to Investment Grade bonds for their opportunity set, leading to a sharp climb in spread at the next segment, so the bond price “falls of a cliff” once the credit quality is in doubt.

Liquidity Coverage Ratio and Net Stable Funding Ratio

An additional concern with liquidity is the funding mismatch between a defaulted obligor in a loan portfolio or structured credit holding, against the liability calls on the business, basically a de-leveraging of the balance sheet during a period of financial stress. The Basel regulatory response to this is to create guidelines for the Liquidity Coverage Ratio mandating a minimum amount of High Quality Liquid Assets in the Fixed Income portfolio to cover one month of outgoings, and the ability to access funding lines and equity issuance through a one year period which is measured by the Net Stable Funding Ratio¹⁸. This is similar to an investment mandate maximum on illiquid assets, where the balance sheet is treated as a funded amount. We will cover this in the liquidation protocol, and also review how this analogy creates a parallel between banks and funded vehicles that perform the same purpose.

Illiquid Assets: Private Credit, Real Estate, and Infrastructure

We have seen how Structured Credit and Financial Bank bonds are basically Private Credit, so there is a strong triadic awareness between all three: they are the same discounted cash flow packaged differently. Most of these are collateralised by Real Estate, so we can further regard Real Estate as Potential Credit: the same way in Physics that we can see how Kinetic Energy comes from Potential Energy. So, fully funded equity holdings of Real Estate, are potential credit. Furthermore, Infrastructure, with its stable earnings, local monopoly position in providing a utility service, can be seen as a potential sovereign bond of the highest credit quality, if the enterprise is unencumbered by any other debt. So in a sense, illiquid assets are potentially liquid credit assets, with the same discounted cash flow model underlying the valuation.

Epilogue on Fixed Income: Lots of Commonality in Risk Factors

As a final comment on Fixed Income, even though the asset class is highly heterogeneous and illiquid, there is a lot of commonality in the risk factors, such that any fixed income asset, including illiquid ones, can be modelled in the same way: based on fixed and fairly predictable cash flows accounting for a yield, and then discounted by a common market funding rate, a sector-based credit spread, and finally an idiosyncratic liquidity spread. All fixed income assets, even private assets, submit to this type of modelling. This indicates that Fixed Income lends itself well to aggregating positions and hedging out the risk of common risk factors. Many of the problems emanate from leverage, where a forced sale is the catalyst of selling into a void.

Liquidation with Single Position Holding

Let us start with an asset exposure that is not hedged that requires a timely liquidation against a market size and trading volume that presents a problem given the position size. From our earlier taxonomy, we look at the market microstructure metrics for Trading Volume, Bid-Ask Spread, and Standard Deviation of market returns for the asset. We change the numéraire to be measured in Position Weight (%) for generality. First, we set up the trade schedule in order not to have a price impact by determining how much of a cut from the trading volume we can take in a set interval. This gives us an Unwind rate per unit of time.

Passive Chase without Crossing the Spread (No Price Impact)

Passive Cut of the Trading Volume Gives us the Chase without Footprint

$$\begin{aligned}\kappa \text{TradingVolume} &= \text{Chase}_\tau \\ \frac{\text{Chase}_\tau}{\text{PositionSize } w_{t_0}} &= \text{LiquidationHorizon } T \\ \text{Unwind}_\tau &= \frac{1}{T}\end{aligned}$$

For example, with a Cut of 10% of Trading Volume, we can Chase 10% of the market daily. If we have a Position Size that represents 100% of the daily Chase, that gives us a Liquidation Horizon of 10 days. We can now express the Chase as a linear decay rate in terms of Time: $\frac{1}{T}$.

Khan Return Function with Linear Volume Weighted Average Price

The trade schedule by following the Chase is linear, with a volume weighted average return function equal to:

$$\begin{aligned}f_t(r_{\pi_T}) &= w_{t_0} \int_0^T \left(1 - \frac{1}{T}\right) dt \\ f_t(r_{\pi_T}) &= w_{t_0} \frac{T}{2}\end{aligned}$$

Thus, in this example, where the Liquidation Horizon is 10 days, the average Time with a Linear Unwind Schedule is 5 days. We will now replace the return function with a risk estimate.

Khan Liquidity Value at Risk Formula Linear Unwind Trade Schedule

If we use this as a return function, and insert a return equal to the market uncertainty parameterised earlier, with a Stressed VaR $\Phi(X < 1\%)$, using $Z_p = -2.33$:

$$f_t(r_{\pi_T}) = w_{t_0} Z_p \sigma_\kappa \sqrt{\frac{T}{2}}$$

We will use this Value at Risk function as one of our optimisation goal functions for unconstrained weights in later sections.

Active Chase with Crossing the Spread (Price Impact)

We now look to cross the spread with an active chase that larger than the neutral cut and thereby create some price impact, taking advantage of the fact that Order Flow is a random variable (Almgren-Chriss model⁴).

Khan-Poisson Model for Random Order Flow

We simplify the more complex mathematics deployed in the Almgren-Chriss (arrival and rebound rate) model with a Poisson distribution that assumes that order flow is independent and arrives at an average rate of λ such that the inter-arrival times are $1/\lambda$, and therefore a continuous time unwind occurs with an exponential decay rate of $e^{-\lambda t}$. This is convex in the second order differentiation, which means that an optimal point will be at time λ_t . If we fit this to the linear unwind by setting the last increment equal in both models, we can parameterise λ_t as:

$$e^{-\lambda_t(T-1)} = \frac{w_{t_0}}{T}$$
$$\lambda_t = \frac{-\ln\left(\frac{w_{t_0}}{T}\right)}{T-1}$$

Extending the previous example, with a passive chase of 10%, and a linear 10-day trading horizon: if we parameterise the exponentially weighted trade schedule, we have an active chase of 22% multiplied by the remaining position.

This rate starts getting less than the linear rate of the Passive Cut after time τ^* , which doesn't take full advantage of the market share available for unwind, so we increase it to the linear rate, for the second half of the model; we will look at this extension after first developing the exponentially weighted return function.

Khan Return Function with Exponential Volume Weighted Average Price

$$f_t(r_{\pi_T}) = w_{t_0} \int_0^{T-1} e^{-\lambda_t t} dt + \frac{w_{t_0}}{T}$$
$$f_t(r_{\pi_T}) = w_{t_0} \left(\frac{1 - e^{-\lambda_t(T-1)}}{\lambda_t} + \frac{1}{T} \right)$$
$$f_t(r_{\pi_T}) = w_{t_0} \left(\frac{1}{\lambda_t} - \frac{1/T}{\lambda_t} + \frac{1}{T} \right)$$

Thus, in this example, where the Liquidation Horizon is 10 days, and where the last term is the last day parameterised as a linear increment, giving an active chase of 22%, we get an exponentially weighted average of 3.5 days in the first two terms:

Khan Tau Function for Exponentially Weighted Average Time to Liquidation

$$\tau = \left(\frac{1}{\lambda_t} - \frac{1/T}{\lambda_t} \right)$$

Khan Weight Function for Residual Liquidation Position

Next, we look at how much of the position is left at time τ :

$$w_{\tau} = w_{t_0} (1 - e^{-\lambda(\tau)})$$

$$\text{Where: } \tau = \left(\frac{1}{\lambda_t} - \frac{1/T}{\lambda_t} \right)$$

In our example, it's 40% of the original position. Notice that this is different from a conventional exponential decay, which would front load even more aggressively, such that 30% is left. Heuristically, we wish to see this as a "half-life" so we can approximate it as around half left from a trader's perspective. Asymptotically, it will converge to a half-life measure, the longer the liquidation horizon.

Khan Change of Mésure (Tau) Methodology

At this point, we apply a linear unwind trading schedule since it will be higher, by adding the linear passive chase term as a rate of time and solve for time by setting to zero:

$$w_{t_0} \frac{1 - e^{-\lambda(T-1)}}{\lambda} - \frac{w_{t_0}}{T} dt = 0$$
$$\Delta t = \frac{T e^{\left(\frac{w_{t_0}}{T} - 1\right)}}{w_{t_0} - 1}$$

In our example, we have a remaining 40%, at a passive cut of 10%, gives another 4 days.

So, the total liquidation horizon is 7.5 days. Note that this alteration does not change the original optimally weighted time, which we retain, we just parameterise the last half to close out in 4 days instead of 6.5 days, so we change the Mésure (change of time horizon)⁵. Let's return to our original return function for a Value at Risk calculation.

Khan Liquidity Value at Risk Formula: Exponential Unwind and Price Impact

If we use the exponentially weighted return function, and insert a return equal to the market uncertainty parameterised earlier, Stressed VaR $\Phi(X < 1\%)$, using $Z_p = -2.33$, with the value Tau:

$$\tau = w_{t_0} \left(\frac{1}{\lambda_t} - \frac{1/T}{\lambda_t} \right)$$

And a Transaction Cost in Bid-Ask Spread for the excess Active Chase in % in returns:

$$TransactionCost_{\%} = w_{t_0} \left(\lambda_t - \frac{1}{T} \right) \frac{Ask_{\%} - Bid_{\%}}{2}$$

Then we have a Khan Liquidity Value at Risk Formula in terms of returns as:

$$f_t(r_{\pi_T}) = w_{t_0} Z_p \sigma_{\kappa} \sqrt{\frac{\tau}{2}} - TransactionCost_{\%}$$

We will use this Liquidity Value at Risk function as our optimisation goal function for constrained weights that are at the upper bound constraint.

Intuition behind the Exponential Unwind Model as Optimal Liquidation

Thus, we have arrived at a combination of shorter market uncertainty with a negative price impact from the transaction cost, which combined optimally reduces the value at risk due to a shorter time to liquidation and taking advantage of random order flow. The intuition of this model shows that we reduce the market uncertainty by reducing the time to liquidation but also that the increased active chase results in an aggressive front-loading of the unwind, such that about half the trades incur a transaction cost. The passive chase is 10%, which passes without price impact, and then the excess active chase amounts to 22%-10% at 12%, which incurs a transaction cost.

Also, we note that the 22% active chase is similar to the common trader wisdom of a 25% maximum market absorption rate. Thus, it is common market practice for active traders to capture similar amounts of the trading volume, when they are looking to aggressively dump positions. As we saw, this comes from parameterising the unwind to the passive chase liquidation horizon, so there is a balance between the two market shares. Additionally, from the excess, it looks like about half of the position is going to active trades. This is interesting from a Bayesian prior perspective, of doing about “half-half” in terms of how we approach the unwind. It has an intuitive appeal. Also, from a game-theoretic perspective, we are not the largest person in the room. A dysfunctional market would surely result in any position larger than this.

Finally, we note that we did not discuss the actual transaction cost, other than earlier when we imposed a Good Deal Bound based on either market uncertainty or availability of offsetting self-financing hedge instruments. There is an implicit assumption that the market-maker or counterparty holds a similar understanding of the trade-off between market uncertainty and price impact. We will cover this after we look at how hedging positions first reduce that market uncertainty and therefore incentivise a lower transaction cost even for an unhedged position such as in this case. We can apply the rational model of a Good Deal Bound versus a Hedged Position to determine the acceptable transaction cost function related to the exposure. If a transaction cost diverges from the Good Deal Bound, we can assume the transaction cost is economically inefficient and thereby opt for the linear unwind.

We will retain both the exponentially weighted and linear unwind models as a framework for the evolution of the liquidity risk management to incorporate a hedging strategy. But first, we will look at how multiple positions and weight constraints in the investment mandate will affect the liquidation. This can be seen as a dynamic programming optimisation of the Value at Risk function against the objective of eliminating an overall position. Upper bound constraints create a slack in the unwind rate for some assets, whereby the constrained asset requires an exponential unwind, while the slack asset can be executed with a linear unwind trade schedule. Thus far, $w_{t_0} = 1$: 100% in one position. We will now have a number of positions, with different chases given different trading volumes, and with upper bound constraints on the weights in order to keep the overall portfolio optimally balanced to its investment mandate.

Liquidation Waterfall : Multiple Holdings, No Constraints

When there are no portfolio constraints for groupings with heterogeneous liquidation periods, we can extend the simple single position liquidation procedure to multiple holdings. This would be the case in a forced sale, or when the fund is gated and under liquidation. The result would be the fastest liquidation. Normally, VaR is not additive, but since we are bringing the risk to zero for the fully liquidated positions, we can add them for an overall VaR liquidation measure. If positions in different liquidation groups are highly correlated, for example all in the same asset class, then this will also bring the overall risk to the lowest level; except when it is a partial liquidation, and there are still investors in the remaining portfolio that are looking to maximise their returns. Overall, this “waterfall approach” supplies a broad risk measure suitable for a liquidity risk report for example for the EMIR Liquidity Stress Testing regulatory framework, for what the fund can achieve in a forced liquidation: all holdings would commence a liquidation at the same time, and the longer ones would remain, cascading into liquidity one by one, hence the term waterfall, for a cascading liquidation, all starting at the same time, then finishing when reaching the liquidation horizon for the respective asset, with a sum of the value at risk functions.

$w_{\pi} = \sum_{i=1}^n w_i$:The weights of a portfolio are the sum of the component weights.

Khan Liquidity Value at Risk Formula with Linear Unwind Trade Schedule

If we use this as a return function, and insert a return equal to the market uncertainty parameterised earlier, Stressed VaR $\Phi(X < 1\%)$, using $Z_p = -2.33$:

$$f_t(r_{\pi_{T_i}}) = \sum_{i=1}^n w_{i,t_0} Z_p \sigma_i \sqrt{\frac{T_i}{2}}$$

Khan Value at Risk Formula with Exponential Unwind and Price Impact

If we use the exponentially weighted return function, and insert a return equal to the market uncertainty parameterised earlier, using $Z_p = -1$, first recalling the values for Tau for each asset:

$$\tau_i = \left(\frac{1}{\lambda_{i,t}} - \frac{1/T_i}{\lambda_{i,t}} \right)$$

And a Transaction Cost in Bid-Ask Spread for the excess Active Chase in % in returns:

$$TransactionCost_{\%i} = w_{i,t_0} \left(\lambda_{i,t} - \frac{1}{T_i} \right) \frac{Ask_{\%i} - Bid_{\%i}}{2}$$

Then we have a Waterfall Liquidity Value at Risk as the summation of the partial weight returns (Stressed VaR $\Phi(X < 1\%)$, $Z_p = -2.33$):

Khan Waterfall Liquidity Value at Risk Formula with Multiple Holdings

$$f_t(r_{\pi_{T_i}}) = \sum_{i=1}^n w_{i,t_0} Z_p \sigma_i \sqrt{\frac{\tau_i}{2}} - TransactionCost_{\%i}$$

Full liquidation but also gradual smaller redemptions can be managed in a Waterfall, similar to Asset Liability Management in a bank, with a top-heavy liquid component acting as Treasury operations. For example, if a fund receives a 10% redemption, then the weights being liquidated are 10% of the total portfolio, and hence 10% of each relative weight, while the investment mandate target is still on the overall weight even if the most illiquid position is not liquidated. We delimit this Waterfall liquidation as a discretionary call of a fund manager in normal market conditions, where a fund manager is equally expecting to receive new subscriptions that will neutralise the redemption. A 10% target for the relatively illiquid group with a 5% tolerance, would be a suitable design allowing for a Waterfall approach. In this case, the liquidation can proceed as in the first section, simply from the liquid part. It is important to maintain hard upper limits for the weights of the illiquid component to signal when a proportional portfolio liquidation should occur.

An example of a liquidity mismatch leading to a fund debacle is when there are no constraints on the illiquid part and an unplanned redemption schedule. For example, the Woodford wind-down in 2019 in UK, where £10B was liquidated, from most liquid to least, left £3B of essentially private equity micro-caps as the only holding left in the fund, ending with losses for the remaining investors and penalties on this leading UK asset manager²³. We will look at how to resolve this with a constrained liquidation in the next section.

Liquidation with Multiple Holdings and Constraints

With the general framework developed, we now look at a liquidation with multiple holdings and constraints. Each separate holding has its own trading volume, unwind rate, and transaction cost vector, but due to constraints in the investment mandate, they cannot be pursued separately: they must be unwound together at a constrained rate which retains the proportion of each asset within the investment mandate target range, and with upper (and lower) bounds on asset sub-groups. Finally, in our previous example with a single position holding, the change in position and the total portfolio is the same. We now need to update our understanding of the weight being liquidated as being a partial weight of the entire portfolio, so $w_i = \Delta w_{\pi} / w_{\pi}$.

Proportional Unwind for Optimal Portfolio and Minimum Tracking Error

While a 10% redemption would not trigger any constraints due to target ranges creating a slack around the constraint, a 25% redemption might hit the upper limit constraint for the most illiquid asset group and consequently create a new unwind schedule for the portfolio. An important reason to pursue a balanced unwind is retaining the optimal asset mix. Each position has a marginal contribution to risk that results in a suitable risk profile, and only a proportional unwind, will result in retaining this risk profile. A typical function to optimise in order to retain this risk profile is the Tracking Error, measured as the square root of the squared deviations from the target weight. This assumes equal liquidity across the Tracking Error deviations, so we will avoid this and instead focus on the Value at Risk functions for each weight.

Proportional Unwind Leads to a Bottleneck

Unfortunately, a proportional unwind essentially means that the portfolio unwind is controlled by the slowest unwind rate of all the assets, which means the most illiquid asset dominates the liquidation even though it is the smallest position. We will look at how reverse stress testing a liquidation and adding design elements to the maximum constraint will enable a safety margin for a partial liquidation of a portfolio without triggering the investment mandate constraint. At the portfolio design stage, illiquid components should be restricted to an amount that allows everyday redemptions without requiring liquidation.

Typically, we will find the crucial constraint to be the maximum percentage holding of the most illiquid asset, which coerces a slowdown of the unwind to the liquidation of this holding. For example, in a Global Fixed Income portfolio, the maximum weight in the most illiquid holding, say Structured Credit, might be set to 10%. While the other positions could be liquidated in a day, the Structured Credit unwind in a stressed market would take a month. This situation is referred to as the “tail wagging the dog”: the small portion of illiquid assets have a disproportionate effect on the overall liquidation horizon of the portfolio forcing all the liquidity into a bottleneck situation.

Liquidity Groupings

It is not necessary to have instrument-level liquidity functions; it is sufficient to have groupings of various securities by similar liquidity. Most investment constraints are on asset type groupings, for example constraints on the weightings in Government Bonds, Corporate Bonds by Credit Ratings Groupings (Investment Grade, High Yield), and Structured Credit (CMO, CLO, CMBS, ABS). We will do the same in terms of understanding the weight constraints: these are not weights in separate securities (although those certainly exists as well, we will consider that a risk concentration problem separate from liquidity), but rather in separate Asset Type Groups with similar liquidity characteristics. Every Asset Class and hence Fund Holding, has a standardised way of grouping assets into liquidity groupings. In Equities, it is “Large Cap”, “Small Cap”, “Emerging Markets”, “Micro Cap”, and “Privately Held”, with the last category usually held in a specialised Private Equity fund but sometimes disingenuously mixed up into “Small Cap” funds, which is a matter of operational due diligence of exactly these types of classifications. The actual trading volume is available from the exchange, so it is possible to create liquidation horizons for each grouping. There are industry standard classifications for these groupings by Asset Class, which are to be recommended.

Proportional Unwind with Lagrange Optimisation on Value at Risk

A proportional unwind can be treated as a dynamic programming optimisation, on the Value at Risk function created, using the Exponentially Weighted for the constrained weight function hitting the upper bound, and the Linear for the unconstrained (“slack”) unbounded weights. This is a fairly trivial numerical exercise which we can delimit outside the main research.

We will instead focus on another approach: to create an analytical framework whereby we can arrive at the same answer from a Lagrange optimisation, that uses the difference in separate singular liquidation half-time τ as the Lagrange multiplier value λ . Hence, the use of λ both for τ and for the Lagrange multiplier λ_i will at first be confusing but later coincide in value (where the Lagrange λ_i is the $\Delta\lambda_{i,c}$ between the unbounded asset and the constrained asset). We will use the same procedures as in the single asset liquidation case, then switch the unwind rate to the constrained asset, with the Lagrange multiplier for each slack asset being the difference between the original λ_i and the constrained $\Delta\lambda_{i,c}$.

One complication which we will overlook that a dynamic programming optimisation would solve explicitly, is the choice between the exponentially weighted and the linear weighted decay rates for the weights operating under conditional constraint from the constrained variable. We will create the optimisation first with all variables under exponential weighting since it is the optimal trading scheme, and then switch the conditionally constrained variables to the linear unwind scheme since it will save the transaction cost and hence be optimal.

Definitions

$\mathbf{w}_\pi = \sum_{i=0}^n \mathbf{w}_i$: The weights of a portfolio are the sum of the component weights. Sometimes an added constraint of $0 < \mathbf{w}_\pi < 1$ is added (mainly to avoid over-optimising corner solutions that would result in infinite leverage and infinite pair-wise long-shorts), for long-only and unlevered, but we will not implement this, since at a later stage we will include self-funded offsetting short hedge positions.

$\Delta\mathbf{w}_\pi/\mathbf{w}_\pi$: A partial portfolio liquidation $\Delta\mathbf{w}_\pi/\mathbf{w}_\pi$ leads to positions w_i to be sold. A practical example would be a CIO pension office SAA rebalancing between Funds, or a Fund redemption for an Asset Manager requiring partial liquidation of the portfolio, or a partial liquidation by an investor to buy into another asset.

$\Delta\mathbf{w}_{i_0} = \mathbf{w}_{i_0} \frac{\Delta\mathbf{w}_{\pi_0}}{\sum_{i=0}^n \mathbf{w}_{i_0}}$: This is different from the single univariate case, where the position was treated as the same, even though it was partial. Here, the partial trade, is separate from the overall held position, because the composition may change, especially if there is “slack” compared to the constraint variable. At the initial time, the reduction in the portfolio is multiplied by the individual position, to give the trade amount for that position which is being optimised.

$B_L < \frac{w_i}{\sum_{i=0}^n w_i} < B_U$: This is a crucial distinction. We are assuming an upper and lower boundary condition on weights in the portfolio; it is the Upper Bound that will be a constraint in the unwind. These weights will be implemented in asset type groups, so there can be a number of assets that constitute the constraint. We will view the granularity of this from the level of the constraint, hence treating all sub-weights as one and the same. Positions below will be proportionately decreased, so will not impinge on any asset-level constraint as a result of this focus. Note that the constraints are on the overall position level, not the trade level, so there may be “slack” in the initial part of the unwind.

Note the time-varying denominator, so the boundary constraint is fixed but the denominator is time-varying. This is not customary for Lagrange optimisations, but necessary to retain the intuition of what the position looks like in terms of weight percentage. Another way is to change the constraint itself. So, a 20% portfolio liquidation means the denominator is 0.8 by the end, and a constraint of 10% would be 8%. But it is highly counter-intuitive and difficult to deploy.

Check Each Unwind Against Boundary Conditions

We check if the proposed portfolio partial liquidation hits any boundary conditions. If even one is hit, all must be affected. For example, if it's a 10% liquidation, and there is a 10% maximum position that's initially at 5%, then there is no effect. It will not be affected until a 50% liquidation. But if it's a 25% liquidation and there is a 20% maximum position that is initially at 18%, then it will be impinged, since it will go to $0.18/0.75 = 24\%$.

Trade the Slack First Until the Constraint is Impinged

The first thing we do is actually bring it to the constraint. Any trade up to then is free. So, we trade out using a Waterfall to the first constrained variable. No other constraint will be impinged because we are now doing a proportionate liquidation, so only one will dominate. We don't necessarily redeem the initial amount to the first-come client; we are just creating a liquidity buffer at this stage that will be distributed *pari passu* to all investors. If it is a complete liquidation, we can do it by reverse waterfall, from the least liquid, and then ascending.

Khan Linear and Exponential Unwind Rates for Each Sub-Group Weight

Now we compute the linear and exponential unwind rates from the Market Size and Trade Volume of each asset sub-group, taking into account stress market parameters if so relevant (as required in a regulatory ESMA Liquidity Stress Test²⁴ report for example) resulting in a different liquidation horizon T_i and unwind rate for each asset sub-group:

$$e^{-\lambda_{i,t}(T_i-1)} = \frac{\Delta w_{i,t_0}}{T_i}$$

$$\lambda_{i,t} = \frac{-\ln\left(\frac{\Delta w_{i,t_0}}{T_i}\right)}{T_i - 1}$$

Find the Constrained Lamda and Apply to Other Asset Liquidity Groups

Next we find the weight under constraint and apply its $\lambda_{i,c,t}$ to the other asset liquidity groups. If possible, apply the $\lambda_{i,c,t}$ as a linear unwind for the slack variables, to avoid any transaction cost.

Hence the decision rule, if $\frac{\Delta w_{i,t_0}}{T_i} < \lambda_{i,c,t}$, then use linear unwind scheme for asset i. If in the presence of many variables, we are unsure of the linear unwind, we can retain the exponential for all without further ado, since they will not imply a trading cost all the same.

Summed Exponential Unwind for Each Sub-Group Weight by Constraint

$$f_t(r_{\pi_T}) = \sum_{i=0}^n \Delta w_{i,t_0} \int_0^{T-1} e^{-\lambda_{i,c,t} t} dt + \frac{\Delta w_{i,c,t_0}}{T_{i_c}}$$

$$f_t(r_{\pi_T}) = \sum_{i=0}^n \Delta w_{i,t_0} \left(\frac{1 - e^{-\lambda_{i,c,t}(T-1)}}{\lambda_{i,c,t}} + \frac{1}{T_{i_c}} \right)$$

$$f_t(r_{\pi_T}) = \sum_{i=0}^n \Delta w_{i,t_0} \left(\frac{1}{\lambda_{i,c,t}} - \frac{1/T_{i_c}}{\lambda_{i,c,t}} + \frac{1}{T_{i_c}} \right)$$

Summed Constrained Optimisation arrives at Lagrange Multipliers

$$f_t(r_{\pi_T}) = \sum_{i=0}^n \left(\frac{1}{\lambda_{i,c,t}} - \frac{1/T_{i_c}}{\lambda_{i,c,t}} + \frac{1}{T_{i_c}} \right) \partial w_{i,t_0} - \sum_{i=0}^n \lambda_{\mathcal{L}_i} - \sum_{i=0}^n \mathcal{B}_i = 0$$

Where $\mathcal{B}$ is a vector containing the Boundary Conditions for each w_i and $\lambda_{\mathcal{L}_i}$ are the Lagrange multipliers that set the constraint to zero against the partial derivatives of w_i .

Under the Karush-Kuhn-Tucker theorem, the significance of the $\sum_{i=0}^n \lambda_{\mathcal{L}_i}$ is that each Lagrange multiplier is solved as the difference in the unconstrained exponential weight minus the constrained exponential weight $\Delta \left(\frac{1}{\lambda_i} - \frac{1}{\lambda_{i,c,t}} \right)$ for each weight: each is the reverse slack of the inefficiency in each unconstrained variable. It is the distance between the optimal decay rate and hence optimal time to reach a half-life position τ . So rather than calculate it, we can simply take the difference in the half life. Since this parameter determines the Value at Risk measure, the risk increases by a factor of the square root of the ratio of the two $\sqrt{\frac{\tau_{i_c}}{\tau_i}}$.

The negative of the slack amount of inefficiency in remaining variables can be the liquid amount that is used to finance any offsetting hedge positions; we will look at this in the next section on hedging for liquidation.

Combined Liquidation Horizon = Constrained Lamda Liquidation Horizon

$$\tau_{i_c} = w_{i_c t_0} \left(\frac{1}{\lambda_{i_c t}} - \frac{1/T_{i_c}}{\lambda_{i_c t}} \right)$$

And a Transaction Cost in Bid-Ask Spread for the excess Active Chase in % in returns:

$$TransactionCost_{\%} = w_{i_c t_0} \left(\lambda_{i_c t} - \frac{1}{T} \right) \frac{Ask_{\%} - Bid_{\%}}{2}$$

Calculate for the Slack variables and see they can be done linearly, in which case no Transaction Cost applies to those.

With Stressed VaR $\Phi(X < 1\%)$, $Z_p = -2.33$, we have a Liquidity Value at Risk in terms of returns as:

Khan Constrained Liquidity Value at Risk for Illiquidity Constraints

$$f_t(r_{\pi_{T_i}}) = \Delta w_{t_0} Z_p \sigma_{\kappa} \sqrt{\frac{\tau_{i_c}}{2}} - TransactionCost_{\%}$$

Which effectively increases the risk on the unconstrained securities by $\sqrt{\frac{\tau_{i_c}}{\tau_i}}$.

Khan Paradox of Constrained Liquidation Bottleneck

A portfolio liquidation with heterogeneous liquidation horizons for assets with bounded constraints leads to two counter-intuitive dynamics that need to be addressed. First, “the tail wagging the dog”: the small portion of illiquid assets have a disproportionate effect on the overall liquidation horizon of the portfolio, otherwise we get a liquidity mismatch bottleneck. Second, “a bank run with first-mover advantage”: where the first investors who post a redemption manage to get the best terms, while the laggards are stuck with illiquid assets. Both of these effects are offsetting, so we look at how to action the first constraint in order to avoid the second. In the final section, we discover that hedges can help to neutralise the illiquid asset groupings, by “hibernating” them such that the asset exposure is dormant by neutralising risk factors with hedges. For small neutral redemptions where new subscriptions are expected, it means more likely it’s a good idea to keep a liquidity buffer, as well as to keep it for making hedges; the liquid portion that is bottlenecked can finance the hedges on the illiquid part.

For the first problem of the illiquidity dominating the entire portfolio, the redemption notice must be longer than τ_{i_c} , the liquidation horizon of the longest instrument; even though it is held in a minority share in the portfolio. Second, we retain any waterfall trade up to the constraint as a liquidity buffer, until the total amount is liquidated. This will be distributed *pari passu*. Such a measure prevents a bank run: every investor needs to participate in the least liquid asset liquidation. Equally it makes sense to keep heterogeneous liquidity profiles separate: a Private Equity fund, a Private Credit fund, not a mixed fund since it will be treated by its worst liquidity profile. It is well advised to define the fund around a single liquidity horizon and make a sizeable commitment to the illiquid asset, since it will dominate the risk profile anyways.

Liquidation with Hedging of Illiquid Constrained Assets

As we have seen in the liquidation of multiple heterogeneous assets with investment mandate constraints, the overall portfolio liquidation is disproportionately dominated by the illiquid constrained assets creating a bottleneck in the overall portfolio unwind rate, leading to a paradox as to why they are in the portfolio in the first place. One way to solve this, which is pervasive in the praxis of market-makers and hedge fund managers but not yet in asset managers and asset owners, is the use of hedging to neutralise unwanted exposures; in this case, the entire asset exposure up until the liquidation. Such an acquisition of hedging instruments should be self-financed from the asset itself, which may constitute a problem for an illiquid asset; we shall see that we can use the excess slack in the unconstrained assets to collateralise the acquisition of a hedge exposure on behalf of the illiquid asset. Finally, if an asset manager feels uncomfortable taking on this task, it can be entrusted to a market-maker as part of their fiduciary responsibilities in a controlled partial liquidation. This insight will lead to a more pervasive use of this more efficient method of timely liquidation, and even from a regulatory perspective, we promote hedging instruments as a liquidity management tool²⁵.

Linear Unwind Acceptable for Hedged Exposures, Given Similar Risk

As we do this, this does change the liquidation procedure of the earlier section: it becomes linear. It also changes the time to liquidation: it becomes the half-way point. What influences this choice is the Value at Risk from the liquidation: it decreases to become the Hedge Basis error term, the part of the portfolio that cannot be hedged with the offsetting asset. The lower this is, the lower will be our risk aversion of the overall market uncertainty. Let's recall the Liquidation Horizon imposed by the liquidity constrained asset on the portfolio:

$$\tau_{i_c} = \left(\frac{1}{\lambda_{i_{ct}}} - \frac{1/T_{i_c}}{\lambda_{i_{ct}}} \right)$$

$$\text{Where: } \Delta w_{\pi_\tau} = 1 - e^{-\lambda_{i_{ct}}(\tau_{i_c})}$$

This is still technically correct. It will produce a Value at Risk which is lower than the linear decay rate Value at Risk. But it will matter less, so there is a level of indifference to choosing the linear unwind, mainly for operational simplicity given the greater number of moving parts and the need to communicate with a trading desk or market-maker to coordinate the unwind. Hence it is easier to implement. For the purposes of simplicity in communication, we will demonstrate the methodology with the linear unwind, with the encouragement to use the exponentially weighted unwind at the discretion of the investment manager should the hedges produce more basis risk, such that the Value at Risk difference between the two makes the Exponential Unwind prevail. Thus, recalling the linear equivalent:

$$\tau_{i_c} = T_{i_c}/2$$

$$\text{Where } \Delta w_{\pi_\tau} = 50 \text{ and the unwind is } 1/T_{i_c}$$

Since the portfolio liquidation is proportional to this constrained liquidation rate, we can use it to determine how much of the portfolio is left at this point. Now that we have a hedged portfolio, and are more indifferent to the market uncertainty, we are also more favourable towards a longer linear unwind instead of aggressively front-loading the liquidation which has a cost.

Hibernating Illiquid Positions during Liquidation with Hedges

We call the act of neutralising the positions during the liquidation horizon “hibernation”. In industry jargon, this is sometimes referred to as “fading out” a position. We determine offsetting hedges for ALL the positions in the liquidating portfolio, including the liquid ones, in order to dispose of all the positions in a proportional wind-down, and hence finance the illiquid positions with the slack of the unconstrained liquid portfolio component.

First, let’s determine what hedges we can use. The ideal hedging instrument would have a correlation of 1, such that we could take an opposing position, in this case an equivalent Buy of a hedge for the Sell of the underlying security and then dispose of them at the slower liquidation rate of the illiquid position. This ideal circumstance does not exist. Instead, we are taking generic, standardised instruments, which offset the risk factor, not the instrument itself. This often means that the risk factor will be slightly divergent to the purpose, resulting in a correlation less than one. Either from duration mismatch, underlying factor mismatch, exchange-traded market microstructure leading to basis risk, or even sovereign mismatch (hedging a SEK Rate Swap with a EUR Rate Swap, as was necessary during October 2008, when there was no quoted market in the SEK during the month). We will include a section covering the correlation boundary conditions for a hedge and ascertain that the correlation can be quite low and still reduce the overall risk of the portfolio liquidation.

Security-Level Risk Models, Aggregating Risk to Portfolio Level

For the determination of the hedge instrument and amount, we need to impose structure on the underlying portfolio by defining the cash flows, risk factors, and valuation dynamics with a risk model or pricing model on the security level. All of these will create basis risk, that we are assuming in exchange for the underlying risk, with the view that they are minute fraction of the original market uncertainty risk. So, for example a government bond can be hedged with a swap rate, although the swap rate may contain systemic liquidity risk itself. A government bond may be hedged with a rate future, although the rate future may have a different duration and hence a duration gap risk. A corporate bond can be hedged with a generic swap and a credit default swap on the market on an index, although the corporate bond may then contain idiosyncratic risk which implies that it defaults while the credit default swap index does not. And equities can be hedged with the market index, with the same risk that the equity contains idiosyncratic risk which the market index does not capture. In all these cases, the underlying hypothesis is that there is a strong correlation between the two that minimises the residual basis, and the price is driven mainly by these risk factors that are captured.

Portfolio-Level Hedges, not Security Level

One further insight is that we don’t have to establish pair-wise hedges: we can do hedging on the portfolio level, across all liquidity groupings. From first having a granular risk model on the security level, we can aggregate them to the portfolio level and implement net hedges on this level. All liquidity groupings will presumably share some basic risk factors, for example interest rates, which can require a common hedge that nets the portfolio sensitivity. Some liquidity groupings may have specific risk factors, which can be hedged separately, for example credit risk for corporate bonds. The hedge term should be for the longest liquidation horizon, and then partially liquidated at the unwind rate in line with the portfolio diminishing in exposure.

Portfolio Value Collateralises the Self-Funding Hedges

A final analysis on the hedge, is that the liquid component of the portfolio is serving as collateral, and therefore we wish to have similar risk attributes between the liquid and illiquid components of the portfolio. This is typically the case by design and intention. The illiquid portion should be higher yielding, with the higher risk coming from the illiquidity premium or embedded leverage, not from markedly different risk factors applying to the securities.

Right Way Risk of Hibernating Hedges

Clearing brokers usually raise red flags for “Wrong Way Risk”, whereby collateral reduces in value in correlation with losses incurred, but offsetting hedging positions lead to “Right Way Risk”. Since the offsetting hedges are by design well correlated with the underlying positions, and taken in opposite directions of the underlying positions, this constitutes “Right Way Risk” in collateralising the profit and loss margin account (P&L) of the Hedges with the underlying liquidating portfolio. During liquidation under market uncertainty, the positions will increase in value when the hedges decrease in value, and the positions will decrease in value when the hedges increase in value, such that the discrepancy is the residual P&L, which is attributed to the Hedge Basis Risk that we were unable to capture with the offsetting hedge positions. However, the underlying position is illiquid, which is never appreciated as collateral. Thus, there is an implicit underlying assumption that the liquid positions, correlate well with the illiquid positions as well, such that they have the same risk attributes in a risk model, only the illiquid positions generate an illiquidity premium for being less available. A good example would be in Credit, where structured credit is more illiquid and earns a higher yield than the more liquid government bond and financial corporate bonds which would serve as collateral for the hedges. One caveat with Right Way Risk is that the hedges in a liquidation will still require acquiring positions in a “One Way Market”, such that for example in a falling market, a short position would have to chase the price downwards. This is similar to the weakness of the portfolio insurance debacle. Still, a position must be liquidated, and thus the hedge is easier to liquidate than the illiquid position, which would incur far higher costs. One can view the hedge for the term of the entire liquidation horizon as an “instant liquidation” of the illiquid position, since it neutralises it until the time that it can be sold.

Sensitivity-Based Hedges

Once we have risk factor models attached to each security, we can aggregate risk factor sensitivities on the portfolio level and net them to zero against hedging instruments. We transition from a correlation-based approach in hedges, to sensitivity-based risk factor models, because by definition identical risk factor models have a correlation of 1, leaving only basis risk as the uncorrelated term. In later sections we’ll look at a toolkit for hedging that transitions from correlation-based hedges at a basic level to risk factor-based hedges, with the understanding that both are imperfect and give rise to basis risk. Examples of sensitivity-based hedges are Duration for Fixed Income, usually in terms Key Rate Bucketing, DTS for Credit against Credit Indices, and Market Beta for Equity against relevant Indices. So, in terms of risk factors, they are perfectly correlated, but then the basis represents idiosyncratic risk associated with the individual positions. With more individual positions, this risk diversifies to zero.

Proportional Linear Unwind with Hedging for Hedge Basis Value at Risk

Employing the same definitions as in the previous section, and the same methodology of setting the liquidation horizon to the constrained security for all unconstrained securities, except linear, we arrive at:

$$\tau_{i_c} = T_{i_c}/2$$

Where $\Delta w_{\pi_\tau} = 50$ and the unwind is $1/T_{i_c}$

Having decomposed the portfolio to risk factors for rates, credit, and equity, we take on hedges for the purpose of netting out the sensitivities, offsetting like-for-like, in interest rate swaps, bond futures, or equity futures and swaps. We only consider delta-1 positions in standardised hedging products because we wish to retain low margin for assuming the position, low transaction costs, and easy of disposal in the market as we unwind the portfolio.

Recalling from the earlier section, that:

$$\sigma_\varepsilon = \sqrt{w^2\sigma_\pi^2 + h^2\sigma_h^2 + 2w\sigma_\pi h\sigma_h\rho_{\pi,h}}$$

$$\text{Setting } h^* = -w_\pi\rho_{\pi,h}\frac{\sigma_\pi}{\sigma_h}$$

$$\sigma_\varepsilon = w_\pi\sigma_\pi\sqrt{1 - \rho_{\pi,h}^2}$$

This can also be expressed as $h^* = -w_\pi\beta_h$ as a Hedge Factor Beta which neutralises the portfolio exposure against the hedge factor. Then the term $1 - \rho_{\pi,h}$ can be understood as the Explained Sum of Squares ESS in an OLS regression, and the σ_ε as the residual error term. To express it more simply: the Basis Risk is the residual P&L of the hedged portfolio.

We take on the hedge at the full portfolio value at the start of the liquidation. We unwind the hedge at the same rate as disposing the portfolio, in a proportionate manner to maintain a net zero factor exposure, retaining a proportional residual basis exposure as the risk. We therefore take on a cost of transactions for the full round-trip of the hedge on the entire portfolio:

$$TransactionCost_{\%} = 2 \cdot \Delta w_{\pi_{t_0}} \frac{Ask_{\%} - Bid_{\%}}{2}$$

Then we have a measure of worst expected returns as Hedged Value at Risk during Liquidation:

Khan Hedged Liquidity Value at Risk : Linear Unwind and Transactions

Stressed VaR $\Phi(X < 1\%), Z_p = -2.33$

$$f_t(r_{\pi_T}) = \Delta w_{t_0} Z_p \sigma_\varepsilon \sqrt{\frac{T_{i_c}}{2}} - TransactionCost_{\%}$$

Which effectively decreases the risk on the portfolio by a factor of: $\frac{\sigma_\varepsilon}{\sigma_\kappa} \sqrt{\frac{T_{i_c}}{2\tau_{i_c}}} \sim \sqrt{1 - \rho_{\pi,h}^2}$

Intuition behind the Hibernation of the Portfolio for Liquidation

Even though the time horizon is longer, this is by far the best Liquidation Methodology of all the ones presented hitherto; Value at Risk is reduced to a fraction of the original market uncertainty, simply by taking on offsetting hedging positions. Even with a relatively poor hedge with a correlation of 0.9, the residual risk will be $\frac{\sigma_\varepsilon}{\sigma_\pi} = \left(\sqrt{1 - \rho_{\pi,h}^2} \right)$, 45% of the original risk, making a hedging strategy a dominant strategy for managing a liquidation, especially in stressed markets.

Moreover, although the transaction costs for the hedge are on the whole portfolio as opposed to only the illiquid portion of the portfolio, hedging instruments tend to be standardised, highly liquid securities with low transaction costs compared to the illiquid components, so they are not like for like; the hedging instrument transactions will certainly be much lower to the point of being almost negligible in comparison, especially if they are exchange-traded futures. The decision to hedge the whole portfolio, should depend on the size of the illiquid portion of the portfolio compared to the cost of hedging the whole portfolio, not just the illiquid portion. This is counter-intuitive but can be understood from the constrained liquidation schedule imposed by the illiquid portion on the whole portfolio, as well as the fact that the liquid portion is funding the acquisition of hedge instruments in a collateral arrangement that will favour liquid securities.

The intuition behind the hedging of the portfolio under liquidation is that the hedges neutralise any market exposure of the portfolio, leaving only idiosyncratic risk associated with the actual portfolio. Since they are uncorrelated with each other and with market factors, the larger the number of positions in the portfolio, the more these idiosyncratic risks diversify to zero at a rate of $1/n$. Fixed Income will have a very high correlation to the hedges, about 0.97, while Equities will not. But equities taken in a diversified portfolio, will also have a very high correlation to the market hedge, approaching 1 in greater numbers.

The idea of adding the hedges, is that it is like an “instant liquidation” of the portfolio, or at least a “hibernation” of the risk factors associated with it, while the portfolio is being sold. There is an added impetus for hibernating the portfolio during liquidation, and that is that it will remove a great deal of uncertainty and therefore give more certainty to investors who are redeeming from a fund. The fact that the ability to hedge exists, may very well make investors more patient and less likely to redeem. A liquidity management technique that benefits investors in general, may be to hibernate an illiquid portfolio during stressed market conditions, even when the goal is not to liquidate it.

A final aspect of the hibernation of a portfolio with a hedge on exposures being equal to an instant liquidation, is that it affects not only the downside, but the upside as well. The upside return of the portfolio will not be higher than the basis risk either. But since the expected idiosyncratic risk is positive, the expected drift will still be positive. We will look at the return characteristics and implied cost of carry of the hedged liquidation portion in the next section.

Cost of Carry of the Market Proxy Hedge Instrument Combined in Portfolio

Portfolio in Geometric Brownian Motion: $dP/P = \mu_\pi dt + \sigma_\pi dW_{i_t}$

Portfolio in Market and Idiosyncratic Terms: $\mu_\pi = \mu_m + \mu_\varepsilon$ and $dW_{i_t} = dW_{m_t} + dW_{\varepsilon_t}$

Market Proxy in Geometric Brownian Motion: $dX/X = \mu_m dt + \sigma_m dW_{m_t}$

Hedge Instrument on Market Proxy: $F = Xe^{-r_t t}$

Partial Derivatives of Hedge Instrument F:

$$dF/F = -r_t X e^{-r_t t}; \quad dF/dX = e^{-r_t t}; \quad d^2F/dX^2 = 0$$

Itô Formula for Stochastic Drift in terms of Underlying Market Proxy:

$$dF = -r_t X e^{-r_t t} dt + \mu_{m_t} X e^{-r_t t} dt + \sigma_{m_t} X e^{-r_t t} dW_{m_t}$$

Replace Underlying with the Hedge Instrument F:

$$dF = (\mu_{m_t} - r_t) F dt + \sigma_{m_t} F dW_{m_t}$$

Stochastic Drift of the Hedge Instrument:

$$\frac{dF}{F} = (\mu_{m_t} - r_t) dt + \sigma_{m_t} dW_{m_t}$$

Short Position Stochastic Drift of the Hedge Instrument:

$$-\frac{dF}{F} = (r_t - \mu_{m_t}) dt + \sigma_t dW_{m_t}$$

Combined Equivalent Hedge and Portfolio in Market and Idiosyncratic Terms and setting P to F:

$$\frac{dP}{P} - \frac{dF}{F} = (\mu_m + \mu_\varepsilon) dt + \sigma_i (dW_{m_t} + dW_{\varepsilon_t}) + (r_t - \mu_m) dt + \sigma_t dW_{m_t}$$

Eliminating the Combined Market Proxy element and Combining Stochastic Process $\sigma_\pi = \sigma_t$:

$$\frac{dP}{P} - \frac{dF}{F} = (\mu_\varepsilon + r_t) dt + \left(\sqrt{1 - \rho_{\pi,h}^2} \right) \sigma_\pi dW_{\varepsilon_t}$$

Having n similar but uncorrelated idiosyncratic positions in the Portfolio:

$$\frac{dP}{P} - \frac{dF}{F} = \left(\frac{\mu_\varepsilon}{n} + r_t \right) dt + \frac{\left(\sqrt{1 - \rho_{\pi,h}^2} \right) \sigma_\pi}{\sqrt{n}} dW_{\varepsilon_t}$$

Thus, in an ideal world, we would have a risk-free portfolio earning the funding rate plus residual return and risk: $\sigma_\varepsilon \sim \sigma_\pi \left(\sqrt{1 - \rho_{\pi,h}^2} \right)$, or $\frac{\sigma_\varepsilon}{\sigma_\pi} \sim \sqrt{1 - \rho_{\pi,h}^2}$ which is the idiosyncratic risk not cancelled by the market proxy. But neither is the residual return cancelled by the hedge proxy carry: the reason for the position is still in the return, it's not being "carried away". Even with a hedge in place, it's still being earned to the investor.

Intuition of Cost of Carry of the Hedged Portfolio Under Liquidation

The idea of an instant liquidation is attractive since it is indeed the objective of the manager. But it affects the upside as well as the downside. The upside, however, is still positive. We are fully funded in the position, without leverage, and then the hedge itself is funded by lending out the position, so it is actually earning the funding rate, while it is shorting the hedge, which is short the market factor. Intuitively, it is a securities lending repurchase agreement, earning the repo rate, and risk-free because it is collateralised. Thus, we have $d\mu_\pi = w(\mu_i - \mu_m + r_t)dt$ as the portfolio return for each of the specific portfolio weights, summing up to the overall residual portfolio return. If each μ_i then decomposes into an idiosyncratic term plus the market term $\mu_\varepsilon + \mu_m$, then we have the residual $d\mu_\pi - df/P = w(\mu_\varepsilon + r_t)dt + \sigma_\varepsilon dW_{\varepsilon_t}$ which makes up σ_ε .

The idiosyncratic risk is that the illiquid holding suffers a loss that is unrelated to a market factor. In practical terms, the more of those, the more unlikely that it will have severe impact during the liquidation. If the portfolio is well diversified and the idiosyncratic terms are weakly correlated, then over n securities $\mu_\varepsilon/n \rightarrow 0$. Thus, in an ideal world, we are left with a risk-neutral portfolio when all weights are considered $d\mu_\pi = (r_t)dt$ and $\sigma_\varepsilon = \frac{\sigma_i}{\sqrt{n}} dW_{\varepsilon_t}$. At the very worst case, there is still right way risk in the construction of the hedge, such that a severe loss from a particular holding will most likely mean a market stress, such that the market factor hedge will mitigate any loss from the particular security. From a portfolio manager perspective of a manager who is reluctant to abandon a position, the investor is able to keep the excess return plus obtain a risk-free funding rate, while getting rid of the market return, which may be perceived as an advantage during the liquidation process. If the illiquid position is earning a yield during the liquidation, the excess return over the market proxy would remain.

Next, we will look at how this answer changes with a Hedging Toolkit for Hibernation during Liquidation, going from the most correlated netting positions to the correlation boundary conditions for the least acceptable diversifying positions, and then some use cases in between. Finally, we will look at risk factor models whereby we can convert the correlation factor to a sensitivity and hedge those instead, with the understanding that it leaves basis risk. Next, we will convert the entire illiquid position to a factor model and see how this behaves. This is particularly useful for holdings in incomplete markets such as private banking books, real estate, and infrastructure. Finally, we will look at how to hedge out FX risk during liquidation, which is something that applies to all the asset classes in varying degrees, so we will look at what the optimal configuration will be for each, and how to deal with the funding mismatch between the ultra-liquid FX hedge that requires daily mark-to-market and the illiquid underlying assets.

Hedging Toolkit for Hibernation during Liquidation

In this section, we will review how to actually do the hedging, from the most acceptable netting positions that carry no risk, to the minimum acceptable boundary condition for a diversifying position that reduces risk, to how to convert correlation into sensitivity-based hedging.

Netting Positions

Netting positions are so perfectly correlated, that they net out and novate, which is called “shredding”. A good example are interest rate swaps: let’s assume they are fully leveraged, with the absolute minimum margin assigned, let’s say 50K for a 10M swap. Let’s say that instead of having forward-starting swaps that are identical and hence novate automatically, we have slightly different swaps, with a few days in between, with the same tenor but rolling forward with gradually different terms, buying and selling, so we’re still net 10M, but now we have gross positions of 1B. This is not only illiquid but moreover unique positions: you cannot go back in history and transact in the same. But you can net them out, with a perfect position that closes out any residual exposures. This is called “compression”: the net exposure left from the group of swaps is transferred into sensitivities, and then a closing offsetting swap nets out these sensitivities, so that the net sensitivity is zero. At that point, the entire slew of contracts is novated, so we were carrying 1B in swaps, fully leveraged on 50K, but with zero risk. In other words, it’s not the leverage that is bothering us, it’s the risk. And in this case, the more leverage, the more we can net it out. This is actually a practical real-life example- it happens all the time.

Essentially, this is what we are trying to do with the hibernation of a portfolio with hedging, so the idea of netting and novation serves as an ideal paradigm of what we are trying to attain.

Minimum Correlation Boundary Conditions for Risk Reduction

The main discomfort with hedging is that it's leverage: we are INCREASING positions in order to DECREASE risk. So, there is something deeply counter-intuitive about it. But it's OFFSETTING positions, so we are netting out exposures, like in the previous example, where it's all leverage, but reduces to be risk-free. So, let's look at what are the minimum correlation boundary conditions for a position to reduce risk while we're actually increasing our positions with leverage. This was first done in a paper by Khan on Currency Hedging⁷. In that particular use case, where the position had to be positive, the minimum correlation boundary condition was negative. Here, where we are allowing a negative position in the hedge, we therefore arrive at the reflective quantity of a position having a minimum positive correlation in order to serve as a minimal hedge. Recalling the original risk, this time set as the variance of the sum of portfolio variance with hedge variance, as basis variance.

$$\sigma_{\varepsilon}^2 = \sigma_{\pi}^2 + \sigma_h^2 = w^2\sigma_{\pi}^2 + h^2\sigma_h^2 + 2w\sigma_{\pi}h\sigma_h\rho_{\pi,h}$$

Then, let's determine the correlation boundary condition whereby $\sigma_{\varepsilon}^2 > \sigma_{\pi}^2 + \sigma_h^2$, which will ascertain the minimum correlation for a hedge to reduce the risk of the portfolio.

$$w^2\sigma_{\pi}^2 > w^2\sigma_{\pi}^2 + h^2\sigma_h^2 + 2w\sigma_{\pi}h\sigma_h\rho_{\pi,h}$$

$$\frac{h^2\sigma_h^2}{2w\sigma_{\pi}h\sigma_h} > \rho_{\pi,h}$$

$$\frac{h\sigma_h}{2w\sigma_{\pi}} > \rho_{\pi,h}$$

Next, we set the risk exposure between the volatility-weighted hedge exposure to be the same as the volatility-weighted asset exposure, except negative (meaning the inequality inverts):

$$-h\sigma_h = w\sigma_{\pi} : \rho_{\pi,h} > \frac{1}{2}$$

Thus, already at a low correlation of 0.5, with an offsetting hedge position that is sized to the same volatility-weighted exposure, we will receive a reduction in the overall risk. Note that it will almost surely be a negative offsetting position; a bond would not offset equity as an incremental position, for example. Only in currencies can we get negative correlations that would involve a diversifying effect, which we'll review in the last section for currency hedging.

An example of a hedge meeting this low correlation boundary, would be a short equity index future to hedge a private credit liquidation. The volatility weighting ratio would be such that the short equity future would be about 40% of the credit exposure. Although not a massive reduction in overall risk, it would have lower risk than an unhedged private credit liquidation, presumably about 75% of the risk. We will look next at how to further improve the hedge.

Khan Optimal Hedge Correlation for Risk Reduction

We've already used the optimal hedge ratio for the basis risk, but let's derive it formally:

$$\sigma_\varepsilon^2 = \sigma_\pi^2 + \sigma_h^2 = w_\pi^2 \sigma_\pi^2 + h^2 \sigma_h^2 + 2w_\pi \sigma_\pi h \sigma_h \rho_{\pi,h}$$

$$\partial \sigma_\varepsilon^2 / \partial h = 2h \sigma_h^2 + 2w_\pi \sigma_\pi \sigma_h \rho_{\pi,h}$$

$$\partial^2 \sigma_\varepsilon^2 / \partial h^2 = 2 \sigma_h^2$$

The second derivative is positive and therefore the function is convex, such that the minimum:

$$2h \sigma_h^2 = -2w_\pi \sigma_\pi \sigma_h \rho_{\pi,h}$$

Khan Optimal Hedge Ratio Weight to Minimise Risk

$$h^* = -w_\pi \rho_{\pi,h} \frac{\sigma_\pi}{\sigma_h}$$

Or equivalently as a Hedge Factor Beta combining the correlation and volatility ratio:

Khan Optimal Hedge Factor Beta to Minimise Risk

$$h^* = -w_\pi \beta_h$$

Notice that we wish the ratio of volatility-weighted hedge exposure to be equal to volatility-weighted asset exposure except negative, but instead of 1, it's equal to the correlation

$$\rho_{\pi,h} = -\frac{h^* \sigma_h}{w_\pi \sigma_\pi}$$

Setting h^* in the original equation, except as the square root for standard deviation:

$$h^* = -w_\pi \rho_{\pi,h} \frac{\sigma_\pi}{\sigma_h}: \sigma_\varepsilon = \sqrt{w_\pi^2 \sigma_\pi^2 + h^2 \sigma_h^2 + 2w_\pi \sigma_\pi h \sigma_h \rho_{\pi,h}}$$

$$\sigma_\varepsilon = \sqrt{w_\pi^2 \sigma_\pi^2 + w_\pi^2 \sigma_\pi^2 \rho_{\pi,h}^2 - 2w_\pi^2 \sigma_\pi^2 \rho_{\pi,h}^2}$$

$$\sigma_\varepsilon = \sqrt{w_\pi^2 \sigma_\pi^2 (1 - \rho_{\pi,h}^2)}$$

Khan Residual Risk from Optimal Hedge Factor applied to Portfolio

$$\sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

With $w_\pi = 100$ the ratio $\frac{\sigma_\varepsilon}{\sigma_\pi}$:

$$\frac{\sigma_\varepsilon}{\sigma_\pi} = \sqrt{1 - \rho_{\pi,h}^2}$$

Such that the residual risk reduces to zero at correlation $\rho_{\pi,h} = 1$.

Khan Time Boundary Conditions to Correlation for Reducing Risk

Now that we have ascertained the correlation boundary conditions, from minimum hedge to perfect hedge, let us determine the time constraints that will affect the chosen hedge, and how this will affect the correlation requirements. Let's solve by backward induction first: what if the hedge requires a LONGER time period for liquidation, and then see how this results in a similar requirement, where the liquidation is the deciding factor in the timespan required.

Let's start with volatility-weighted exposures in two instruments, such that $\sigma_1 = \sigma_2$ and $w_1 = -w_2$, so they might be perfectly hedged with a correlation of 1. Now let's assume one has a liquidation horizon of T_s and the other has T_L , for a long and short horizon respectively. Let's assume the longer one is a hedge, that we wouldn't have otherwise. Then, we have two portfolios, with or without the hedge, where:

$$w_1^2 T_s \sigma_1^2 > w_1^2 T_L \sigma_1^2 + w_2^2 T_L \sigma_2^2 + 2\rho w_1 \sqrt{T_L} \sigma_1 w_2 \sqrt{T_L} \sigma_2$$

$$w_1^2 T_s \sigma_1^2 > w_1^2 T_L \sigma_1^2 + w_1^2 T_L \sigma_1^2 - 2w_1^2 \rho T_L \sigma_1^2$$

$$\frac{T_s}{2T_L} > 1 - \rho$$

Khan Minimum Correlation Formula for Hedge Increasing Liquidation Time

$$\frac{T_L - \frac{1}{2}T_s}{T_L} < \rho$$

So, the correlation has to be at least the ratio of the incremental time added over the longer time horizon to be a good hedge for the shorter time horizon position.

We can turn this around and apply it to the longer time horizon in general, whereby we will accept the longer time horizon for a hedge, for the shorter liquidation instruments. So, let's say we have a 3-month liquidation for one instrument, and a 1-week liquidation for another, so comparing 60 to 5 (trading) days, then we would need a $115/120=0.96$ correlation in the hedge to justify the longer time period.

We can now use this decision rule for the shorter time period instruments, holding the entire liquidation portfolio for a longer period, while we dispose of the illiquid assets. Thus, for example for a fixed income portfolio, that contains Government Bonds with a one-week liquidation horizon, and a smaller portion of Structured Credit with a three-month liquidation horizon, we would retain the Government Bonds for a longer horizon if we can find a hedge instrument with a 0.96 correlation. Finally, since the correlation boundary requirement is quite high, it makes sense to look at assets which have the same risk sensitivities, since in those cases the model suggests a correlation of 1, but the underlying risk factors will have some degree of divergence.

Corollary: Correlation Boundary for Portfolio Margining with Exchanges

A practical example for a longer time horizon on a hedge is in portfolio margining of OTC (Inter-Bank Dealer) positions versus ETD (Exchange-Traded) futures: exchanges generally require a 2-day margin, whereas OTC requires a 5-day margin. The time boundary condition provides a clear correlation at which the risk is less by placing the combined position in the longer time horizon margin account even though the liquidation is longer. In our general framework, we only assumed the hedge in order to reduce risk; in this case, we have two different margin accounts, with different time periods, that can net out if assigned together in the longer margin account. Let's look at the dynamics using the same assumptions as in the earlier section, with volatility-weighted exposures in two instruments, such that $\sigma_1 = \sigma_2$ and $w_1 = -w_2$, so they might be perfectly hedged with a correlation of 1.

$$w_1^2 T_S \sigma_1^2 + w_2^2 T_L \sigma_2^2 > w_1^2 T_L \sigma_1^2 + w_2^2 T_L \sigma_2^2 + 2\rho w_1 \sqrt{T_L} \sigma_1 w_2 \sqrt{T_L} \sigma_2$$
$$T_S + T_L > w_1^2 T_L \sigma_1^2 + w_2^2 T_L \sigma_2^2 + 2\rho w_1 \sqrt{T_L} \sigma_1 w_2 \sqrt{T_L} \sigma_2$$

Khan Minimum Correlation Exchange-Cleared Portfolio Margining Formula

$$\frac{1}{2} \left(\frac{T_L - T_S}{T_L} \right) < \rho$$

Thus, in this corollary case, where we already have netting positions just holding them in different risk margin accounts by liquidation time horizons, the correlation requirement is low. We have assumed a negative weight relationship between the two assets to make sure that they net out, and in this case, we are willing to accept the longer horizon even for quite diverse assets. In the earlier example of OTC at 5-day, and ETD at 2-day, we are willing to portfolio margin all assets at the 5-day horizon, so long as the correlation is greater than $.5(3/5) = 0.3$. Thus, in any case where the underlying assets are in the same asset class, it makes sense to combine the OTC instruments with the ETD instruments at the longer time horizon, in order to net out the positions and considerably reduce the risk margining, even by 90% in spite of the longer risk margining time horizon. An example might be holding EUR STIRs and a SEK IRS Swap that would net out; it would make sense to retain them in the longer time horizon account. A more extreme example would be a ZAR IRS Swap against GBP STIR Futures, again it would make sense to retain them in the same account, in spite of the longer liquidation time horizon and the relatively higher risk margining imposed by that time horizon; they would effectively net out and hence justify the longer liquidation time.

Sensitivity Based Hedging for Asset Classes

Now that we have determined the high correlation requirements for hedging instruments, we can be confident to arrive at this level if we have risk models based on the same risk factors. Ideally, these would have a correlation of 1. The difference would be that the risk factors would be slightly different in terms of time or idiosyncratic risk, hence creating a basis spread. An asset class in general, is a good idea for a hedge grouping: Equity, Sovereign and Inter-Bank Funding Rates, and Corporate Credit. Some illiquid asset groups may be hybrids of these. The asset risk sensitivity to the factor would be our measure of risk, and by equating the hedge sensitivity to neutralise the risk, effectively equating the volatilities of the asset and hedge.

Khan Optimal Hedge Ratio and Hedge Factor Beta applied to Portfolio

Let's recall the optimal hedge ratio that minimizes risk for a portfolio:

$$\text{Setting } h^* = -w_\pi \rho \frac{\sigma_\pi}{\sigma_h}$$

Or equivalently, $h^* = -w_\pi \beta_h$ as Hedge Factor Beta.

Khan Residual Risk Formula from Optimal Hedge Ratio applied to Portfolio

$$\sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

With $w_\pi = 100$ the ratio $\frac{\sigma_\varepsilon}{\sigma_\pi}$:

$$\frac{\sigma_\varepsilon}{\sigma_\pi} = \sqrt{1 - \rho_{\pi,h}^2}$$

Such that the residual risk reduces to zero at correlation $\rho_{\pi,h} = 1$.

So quite clearly, we are trying to select a hedge that has a correlation of 1, and measuring this effect as sensitivity simply means the ratio of the volatilities, and the weighting of each is simply the volatility-weighted balance between the two, $h^* \sigma_h = -w_\pi \sigma_\pi$.

The residual risk as seen from this perspective, is quite simply the lack of correlation between the assets and the hedges. A quick glance at what else could be done: we could have a Principal Components Analysis for example, of uncorrelated factors affecting the portfolio; or we could have a Factor Analysis that broke down the Market factor into other sub-groups. The problem with this, is that these analytics would not map cleanly to a hedge instrument. Here, the factor and the traded hedge are one and the same. We want to keep it that way. By all means we can dig deeper into the underlying factors, especially for understanding the idiosyncratic risk as maybe a residual factor, or a residual principal component. But ultimately, we appreciate the correlation between the portfolio, even in sub-groups of assets, and the hedge itself, and we apply our judgment to ensure that it is a hedge that corresponds to the same factor model and ideally would be perfectly correlated with the underlying assets.

If we create a quick taxonomy applying to the major asset classes of Equity, Sovereign and Inter-Bank Rates (Rates), and Corporate Credit (Credit), we have a clear understanding of how the correlation works its way into the sensitivity measure for the asset class.

Equity: Market Beta for the Hedge Measure

$h^* = -w_\pi \beta_h$, where the Hedge Factor Beta is the Market Beta. The appropriate hedge instrument is the relevant Equity Index Future to which the stock belongs. This is quite a clear correlation-based sensitivity that doesn't need much explanation. The main discussion would be why it's so imperfect as a hedge. For a single stock, the correlation between the underlying and the market hedge is quite low, highlighting the preponderance of idiosyncratic risk on a single stock. But bear in mind, by definition the market index is the basket of stocks, so it will be one for the composition of the index. Thus, the correlation quickly converges to one.

Khan Residual Risk from Optimal Hedge Factor applied to Portfolio

With the Beta hedge, the error term is effectively the Residual Error Sum of Squares (RSS) in a linear regression, and the Explained Sum of Squares over the Total, is the correlation squared. So, we are left with the standard error:

$$\sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

With $w_\pi = 100$ the ratio $\frac{\sigma_\varepsilon}{\sigma_\pi}$:

$$\frac{\sigma_\varepsilon}{\sigma_\pi} = \sqrt{1 - \rho_{\pi,h}^2}$$

Such that the residual risk reduces to zero at correlation $\rho_{\pi,h} = 1$.

If we recall the calculation earlier regarding decomposing an asset return into a market proxy and idiosyncratic risk, then we have a clear indication of how much σ_ε will dominate the overall relationship. Also, if we recall the portfolio effect, it will quickly tend to zero at roughly 30 securities. So while we do a bad job at the security level, we are able to diversify away the idiosyncratic risk quite well once we have a larger portfolio of assets that we are hedging.

In terms of temporal correlation boundary, Equity tends to be the most liquid for the stocks with the highest market capitalisation, but also have the highest volatility, so therefore still requiring a hedge for any period of time that they are held. With a well diversified portfolio, the volatility should converge to the market index as a good sign that the correlation between the portfolio and the market index is in line, leaving a Beta of 1. For Private Equity, this tends to act as leveraged market index, so therefore a higher Beta than the market index would be required to hedge it, say a Beta of 1.2.

So, the appropriate hedge will be the Beta of the portfolio to the Market Index, and the amounts will be the equivalent notionals. So 1M in the portfolio against 1M notional of the Market Index future, with a Beta of 1. If the Beta is different from 1, there is a strong chance that the correlation is imperfect, and therefore subject to analysis on a correlation boundary perspective, as incomplete.

Again, we can quickly look at why we don't use other analysis, for example PCA or Multi-Factor regressions. The answer is that we want to have a regression against the hedge, and the best hedge is usually the relevant equity market index future. If we wish, we can analyse both the hedge and the portfolio in terms of PCA or Multi-Factor, in order to understand how the residual might load up on the various remainder from a divergence in these factors. Usually, however, this will be obvious to the portfolio manager, who will have clear characteristics in the underlying investments that differ from the market hedge that cause the hedge basis risk.

Sovereign and Inter-Bank Funding Rates: Duration as the Hedge

Government bonds are best hedged with inter-bank funding swap rates, since they are collateralised and cleared, so they can be treated as free from counterparty risk. We will treat them in the same risk model, with the understanding that there is a basis risk between them, which will show in the residual risk. That said, the correlation between the two is by far the highest from all the asset classes; so much so, that we can often treat the correlation as 1, and just look at the Duration measure as the sensitivity. Let's delve into the Duration measure, and what we can do with that.

We will use the Khan Par Yield Duration as an effective measure of the first order rate sensitivity of a bond, both for Rates and Credit, in the various adaptations⁵. Let's recall what it looks like:

Khan Par Duration Formula for Yields and Spreads (Khan 2016⁶)

Let's look at it as a single measure, just for intuition. So, with the Yield to Maturity y_T :

$$\text{Khan Par YTM Duration } D_{y_T} : dP_{\%T} = \frac{e^{-y_T T} - 1}{y_T} dy_T$$

And then let's decompose into Rate and Spread, with the Yield to Maturity $y_T = r_T + s$:

$$\text{Khan Par Rate Duration } D_{r_T} : dP_{\%T} = \frac{e^{-(r_T + s)T} - 1}{(r_T + s)} dr_T$$

$$\text{Khan Par Spread Duration } D_s : dP_{\%T} = \frac{e^{-(r_T + s)T} - 1}{(r_T + s)} ds$$

Thus, we have a SINGLE Khan Par Yield Duration measure based on the Yield to Maturity, identical to the earlier derivations, that we then can apply to changes in Spread and changes in the underlying Rate. To understand the intuition, it is a single yield measure, YTM, that can then be further analysed by breaking it down into a baseline swap rate and a spread, both of which have an identical effect on the overall price change. We do this break-down in order to apply proxy data.

Spread is the Basis Spread

The spread is our Basis Spread discrepancy between the underlying rate decomposed into the idiosyncratic spread amount and the underlying rate amount, compared to the hedge. The measure is the same, but the discrepancy is a fraction of the original base rate. This will translate directly into the VaR function savings, such that the Duration times Spread is what is left as the residual P/L.

A well-known weakness of this model is that the par yield is static across the whole tenor of the bond, which will especially cause trouble if the hedge and the asset are of different tenors and hence have undue basis risk. We can group the Duration by Key Rates to mitigate this effect. Usually a short-term (0-2Y), medium-term (2Y-7Y), and long-term (7Y+) bucketing will suffice, with the understanding that 15Y+ has its own habitat and will create basis risk.

Khan Marginal Key Rate Par Durations

$$Khan\ Marginal\ Key\ Rate\ Par\ Duration\ \Delta D_{\Delta t} : \Delta P_{\Delta t} / \Delta r_{\Delta t}$$

$$\begin{aligned} dD_{\Delta t} &= dP_{T+\Delta t} / dr_{T+\Delta t} - dP_T / dr_T \\ &= \frac{e^{-r_{T+\Delta t}T+\Delta t} - 1}{r_{T+\Delta t}} - \frac{e^{-r_T T} - 1}{r_T} \end{aligned}$$

Hence, any Marginal Key Rate Par Duration simply by knowing two Par Rates (Yield to Maturities) and two Tenors (Maturities), no further information is required.

Khan Key Rate Spread Duration Formula

$$Khan\ Key\ Rate\ Spread\ Duration : dP_{\%T} = \sum_{t=0}^T \Delta D_{\Delta t} ds_{\Delta t}$$

DV01 Key Rate Spread Duration Formula

Sometimes the Key Rate Duration will be expressed as a DV01 notional, to create a “Hedge Ladder” across key rates, and thereby neutralise the absolute amount.

$$Key\ Rate\ Spread\ DV01 : Notional * dP_{\%T} = Notional \sum_{t=0}^T \Delta D_{\Delta t} ds_{\Delta t}$$

Price Change with Khan Par Duration for P&L Explain of Basis Spread

We can then use these sensitivities for our P&L Explain between Asset and Hedge, where the residual spread will still be the Duration measure, but by the discrepancy spread amount:

Asset – Hedge P&L Explain with the same Tenor:

$$dP_{\%T} = \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} dr_T + \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds - \frac{e^{-r_T T} - 1}{r_T} dr_T$$

So that we are left with the P&L Explain of the Khan Basis Risk Spread:

$$dP_{\%T} = \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds$$

And this is what propagates into the VaR risk function that is our residual risk after the hedge. To use different tenors for the asset and the hedge, we simply multiply each position by the Duration of the asset.

Duration Matching for Minimum Spread of the Asset and Hedge

In the earlier case, we have the same Duration measure for the Asset and Hedge, but we would set the measure as equivalent by Duration matching the value. The hedge has one Duration, and the asset portfolio another; we set them equal by cross-multiplying each by the Duration of the other, such that the cross-product is equal. When applied to the Hedge amount, this gives a ratio that sets the volatility ratio to one. We accomplish having the minimum spread by setting a Duration Neutral Hedge portfolio

Khan Duration Matched Hedge Measure

$$\rho \sim 1: h^* \frac{D_{s_T} \sigma_{r_T}}{D_{r_T} \sigma_{s_T}} = -w_\pi \rho$$

Where there's a strong assumption that the rates are highly correlated, close to 1 for Duration Matching to replace Correlation. Notice that the volatility is the underlying hedge rates and portfolio rates, not the portfolio return. Also, the Hedge amount is multiplied by the Duration of the Asset Portfolio, divided by the Hedge Duration, such that the duration-weighted volatility ratio is one. $\frac{D_{s_T} \sigma_\pi}{D_{r_T} \sigma_h} = 1$, and ideally $\rho = 1$.

Khan Duration Matched Hedge Measure by Key Rates

$$\sum_{t=0}^T h^* \frac{\Delta D_{s_{\Delta t}} \sigma_{r_{\Delta T}}}{\Delta D_{r_{\Delta T}} \sigma_{\Delta s_t}} = -w_\pi \rho$$

This short-cut of duration matching is not always a good assumption, so it requires discretion on the hedge characteristics. For example, an illiquid “Off-The-Run” government bond at 28 years, will have its own particular dynamics compared to a liquid “Cheapest to Deliver” 6-year government bond of the 10Y Bond Future. In this case, we can use the Key Rate Duration for partial hedges across several tenors, to account for the “Slope” in the discrepancy. The Delta of the Duration means that it is actually two instruments, long and short two Hedges. There are spread swaps that can incorporate this effect, for example a single 30Y-10Y Swap.

Khan Residual Risk Duration Formula from Optimal Hedge to Portfolio

With the Duration hedge, where we are making a strong assumption on the correlation being close to one, our interpretation is different. The error term is effectively the basis spread between the hedge and the yield. So, we are left with the standard error:

$$\sigma_{\varepsilon} (s - r_T) = w_{\pi} dP_{\%T} = w_{\pi} \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds$$

With $w_{\pi} = 100$ the ratio $\frac{\sigma_{\varepsilon}}{\sigma_{\pi}}$:

$$\frac{\sigma_{\varepsilon}}{\sigma_{\pi}} = \frac{\frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds}{\frac{e^{-(r_T+s)T} - 1}{(r_T + s)} dr}$$

Such that the residual risk doesn't reduce to zero at correlation $\rho_{\pi,h} = 1$; we already assume it is 1. Instead, it reduces to zero when $s - r_T = 0$. The reason it works, is that the spread is always implied from the yield y , so it is a mark-to-market which implies the spread.

Alternative Correlation Hedging in Rates: PCA and LASSO

There are two main alternative correlation hedges deployed in market-neutral rates portfolios, PCA and LASSO regression. Neither of these are bad ideas; they are used by market-makers and in active market-neutral rates portfolios to optimise the returns. Both map against actual hedges, so actual instruments that can be deployed in a hedge portfolio. And both are to introduce real basis spread concerns, mainly that related to the “Slope” factor, 10Y-2Y, or the “Tilt”, which is the extreme volatility in the 5Y compared to other tenors, the “Belly of the Beast”. About 95% of the volatility can be explained by the Duration, but that still leaves 5% that is related to either the Slope or the Tilt, and which is present in the residual Hedge Basis Spread of the hedged portfolio returns. This is quite clear if you are hedging a 30Y bond portfolio with a 10Y swap: you still have a 30Y-10Y Slope factor that is carried into the residual risk. Thus, let’s look at these refinements, but still delimit them as out of scope although they deserve further study. They are mainly of concern for highly leveraged portfolios where these effects are magnified.

The first strong contender is Principal Components Analysis decomposition of a rates portfolio into Duration, Slope, and Tilt, and then hedging of each. This can be mapped against hedging factors, so it is a strong analytical framework. If there is a large slope factor in the basis hedge error, then it could be a good idea to adjust the hedge in the aftermath of such an analysis. The PCA serves as a proxy for a Multi-Factor regression framework, so let us look at that in particular, since it maps to actual hedges: the first Beta is Duration, same as always, the second Beta is “Slope”, which from our Key Rate Duration example is the Delta of Duration between two key rates, and the third Beta is “Tilt”, which is the convexity of Duration, as measured from the midpoint, we can see it as either long or short Gamma.

$$\sum h^* = -w_\pi \beta_D - w_\pi \beta_{\Delta D} - w_\pi \beta_{\Gamma D}$$

So, for example the first hedge component would be a 10Y Swap, the second hedge component would be a 10Y-2Y Swap Spread, and the third hedge component, would be a Fly Long 10Y-5Y, Short 5Y-2Y. We can easily replicate this result with the earlier Key Rate Duration Matching, which would incorporate the first two terms. The last component of Gamma mainly relates to leveraged portfolios; we refer to the Khan Interest Rate Calculus⁵ for this analytic.

The second method is a hedge selection by a LASSO regression, which loads up a hedge on several candidate bonds. For example, a German bond might be hedged with a combination of German, French, and Italian bond futures by look-through to the cheapest to deliver bonds of each. This can simplify to a linear multi-factor regression with Lagrange constraints as:

$$\sum h^* = -w_\pi \beta_{1D} - w_\pi \beta_{2D} - w_\pi \beta_{3D} - \sum_{i=0}^n \lambda_i$$

So, we find a cheapest-to-deliver blend for the hedge, made up of several components. This maximises yield to a market-maker, so we will delimit this approach as unsuitable for a liquidation. It will tend to be a constrained optimisation towards riskier bonds: for example, it tries to earn the carry between the Italian bond versus the German; thus, instead of minimising the spread, it maximises the spread in the search for yield. This method is used in Credit as well.

Credit Risk: DTS (Duration Times Spread)

The DTS²⁶ analytic emerges directly from the Spread Duration and the Market Spread as discussed earlier, so it is analogous to the study in Rates. DTS is widely used as an aggregate exposure figure, often to compare an active bond portfolio to a benchmark, for example keeping a target Weighted DTS of a smaller active bond portfolio in line with a benchmark portfolio.

That said, it can be accomplished simply by the Spread Duration figure. It is less important that the actual Spread is in line with a benchmark. It is not straightforward to hedge a credit benchmark, and the main reason we are looking to hedge a liquidation portfolio, will tend to be because of illiquid credit. We will look at Credit Default Swaps in the next section.

DTS Duration Times Spread with the Khan Par Spread Duration Formula

Recall the Khan Par Spread Duration Formula:

$$\text{Khan Par Spread Duration } D_s : dP_{\%T} = \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds$$

Then Khan DTS Par Yield Duration Times Spread is:

$$\text{Khan Par Duration Times Spread } DTS_s : DTS = \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} s$$

And the change in price from the hedge market proxy h :

$$dP_{\%T} = DTS \left(\partial s_h / s_h \right) = \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} s \left(\partial s_h / s_h \right)$$

Where s is the credit spread, s_h is the proxy hedge, and the calculation is analogous to calculating a portfolio return from a Market Beta and a Market Return. Notice that it's a calculation on relative spread return. The DTS is the Beta, and the proxy benchmark spread drives the return. This is considered useful in a corporate bond portfolio with heterogeneous durations, spreads, and an overall paucity of data requiring proxy.

Khan Duration Matched Hedge Measure for Credit by DTSS (DTS*Sigma)

$$h^* \frac{D_{s_T} s \sigma_{s_h}}{D_{s_h} s_h \sigma_{s_T}} = -w_\pi \rho$$

To make the figure work for a Duration Matched Hedge, we need to factor in Sigma, so that it's DTSS. Quite frankly, it is still Duration times Sigma, because the Spread term cancels out.

Credit Default Swaps

The best hedge class of swaps for Credit is Credit Default Swap (CDS), which segues into discussions of sovereign and corporate credit in general, since the spread calculated for default can be deployed by proxy in those areas as well. Again, we will brush over some of the intricacies of the market conventions in the Upfront Premium, which have important implications in the Carry and Roll, to focus more on the Par Yield Spread that is the result of the Swap. Here, in the view of a Bank, we receive a fixing k as a premium for a payout D of the amount of the Notional (usually set to a Principal outlay) of a reference bond, either Corporate or Sovereign, with a Credit Event defined as a Default, Restructuring, or Rating Change. The Tenor is usually 5 years, in line with a typical corporate bond.

$$Price_{CreditSpread_T} = Notional \int_{t=0}^T (s - 1 : D_t) e^{-r_t t} dt$$

$$\text{Where: } f_t = dr_t$$

$$D_t = \text{Credit Default Event paying Principal Notional 1}$$

The Par Spread is referred to as the Credit Spread, and we can calculate the Spread Duration as we did earlier. Again, this is usually expressed in terms of an Upfront Premium and then a Spread component as market convention, but for now we will express it as a Par Spread s , as the two concepts are interchangeable on call.

$$\text{Khan Par CDS Spread Duration } D_T : dP_{\%T} = \frac{e^{-r_T T} - 1}{s_h} ds_T$$

CS01: Notional Multiplied by Khan Par Spread Duration for P/L

$$\text{Credit Default Swap P\&L: CS01} = Notional \frac{e^{-r_T T} - 1}{s_h} ds_T$$

Credit Spreads: P&L Explain with Khan Par CDS Spread Duration

$$dP_{\%T} = \frac{e^{-r_T T} - 1}{s_h} ds_T + (s_t)dt + ds_t$$

Where the first term is change in Spread, the second is Carry, and the third is Roll, seen from a Receive Spread, Pay Default perspective. Normally investors will be holding the reverse position as a hedge to a corporate exposure, and therefore the negative of the above.

Khan Duration Matched Hedge Measure for CDS

$$h^* \frac{D_{s_t} \sigma_{s_h}}{D_{s_h} \sigma_{s_t}} = -w_{\pi} \rho$$

There are two different liquid CDS Index products: Investment Grade and High Yield, most likely the hedge would be on the latter, since the underlying asset will be more illiquid.

Khan Residual Risk Spread Duration Formula from Optimal Hedge Portfolio

With the Credit Spread hedge, again the error term is effectively the basis spread between the hedge and the yield. So, we are left with the standard error:

$$\sigma_{\varepsilon} (s - r_T) = w_{\pi} dP_{\%T} = w_{\pi} \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds$$

With $w_{\pi} = 100$ the ratio $\frac{\sigma_{\varepsilon}}{\sigma_{\pi}} : \frac{\sigma_{\varepsilon}}{\sigma_{\pi}} = \frac{\frac{e^{-(r_T+s)T} - 1}{(r_T+s)} ds}{\frac{e^{-(r_T+s)T} - 1}{(r_T+s)} dr}$

Such that the residual risk doesn't reduce to zero at correlation $\rho_{\pi,h} = 1$; we already assume it is 1. Instead, it reduces to zero when $s - r_T = 0$. The reason it works, is that the spread is always implied from the yield y , so it is a mark-to-market which implies the spread.

Currency Hedging

An added risk factor when investing in assets internationally, is the currency exposure, which is often fully hedged. As a risk measure, the annualised volatility of FX is an order of magnitude higher than the annualised volatility of Fixed Income, where it can be shown by the correlation boundary condition that the optimal hedge ratio in this case is a fully hedged Fixed Income portfolio⁷. In this case, there is a carry on the FX forwards, equal to $r_d - r_f$. The paradox for a Fixed Income portfolio, of course, is that we are looking to earn a foreign yield, and yet the price is the foreign money market yield. The implication is that the Fixed Income exposure must have a long end yield that is higher not only than the domestic yield, but higher relative to the foreign money market yield for the Fixed Income exposure to be of interest.

There is one situation, where we might not wish to have a fully hedged Fixed Income exposure, and that is in high-yielding Emerging Markets Sovereign Bonds. In this case, the negative carry on the hedge will render the investment uneconomical. In these cases, the volatility of the Fixed Income reaches levels similar to the FX itself. The optimal currency hedge ratio in these conditions, can be expressed by the Khan Optimal Currency Hedge Ratio with Carry:

Khan Optimal Currency Hedge Ratio with Carry (Khan 2025⁷)

$$\kappa = \frac{\sigma_A}{\sigma_X}; h^* = 1 + \kappa \left(\rho_{A,X} + \frac{(r_d - r_f)}{\sigma_X} \right)$$

Where σ_A is the volatility of the Fixed Income Asset, and σ_X is the volatility of the Currency return, κ is the Kappa ratio of the two, and $r_d - r_f$ is the Carry, all annualised. Here we can see that the hedge ratio h^* will be 100%, unless there is a high ratio κ of the asset to the currency risk, a negative correlation, or a high negative carry to the hedge.

Khan Asset Volatility Formula Derived from Khan Par Yield Duration

$$\sigma_A = \left(\sqrt{\frac{T_f}{\tau}} \right) \frac{e^{-y_T T} - 1}{y_T} d_\tau y_T$$

the first term is simply the annualisation factor, the second term is the Khan Par Yield Duration, and the last term is the first difference in the foreign par rate of a Tenor T over an interval τ (for example a daily first difference).

Let's look at an example with a Brazilian bond investment. So a 10 year Brazilian bond, with a yield of 10%, and a Khan Par Duration of 6, with a daily standard deviation of 0.15%, gives us a rate volatility of 14.5%. The FX Vol is about 15%, so this gives us a κ ratio of about 1.

The correlation of the USDBRL with the local bond rate is positive, around 0.3, which is typical for EM investments from anchor currencies such as USD due to one-way fund flows, so we should expect the hedge ratio to be 100%. But, the carry component is so high, that we need to adjust for it. With a base rate in the USD of 5%, compared to 15% money market rate in BRL, we are looking at -10% in Carry, which adjusted by the FX Vol of 15%, gives us -2/3. So, even with a positive correlation resulting in a 130% hedge, if we adjust for the -66% from the Carry, we arrive at a 50% Optimal Currency Hedge Ratio, for a Fixed Income investment.

There is one more situation where we might wish to have an unhedged FX risk component in a Fixed Income exposure, and that is in highly illiquid Private Credit. In those cases, the liquidation horizon will effectively increase the volatility of the underlying by the liquidation horizon $\sqrt{\tau}$, and hence increase the Kappa ratio, whereby if there is a negative correlation for investing in the foreign market, for example a GBP investor in the USD, then again it will result in an 50% Optimal Currency Hedge Ratio, this time not from the Carry, but rather from the higher liquidation horizon, the high Kappa, and the baseline negative currency correlation.

Thus for the final performance attribution calculations, we need to account for the h^* ratio, whereby the $w\%$ exposure to the international bonds, will have $h^*w\%$ of the Carry element $(r_d - r_f)$ to hedge back into the Base Currency, and $(1-h^*)w\%$ of the FX Return element r_x which is the unhedged FX risk component for the local foreign exposure. We will leave this out, because it is an unusual example, and the exception rather than the rule. But bear in mind that the hedging costs should be included in the Performance Attribution, as part of the Theta Carry term. For now, that is left out, but you can assume that the dr term in the final Carry component has effectively accounted for these two risk factors in the international exposure $w\%$: $(1-h^*)w\%$ and $h^*w\%$ FX Hedge Carry.

Hedging Illiquid Private Assets using a Hybrid Approach

Highly illiquid private assets comprise Private Equity, Private Credit, Real Estate, and Infrastructure. These are privately held businesses, direct loans, property, and utility fixed plants, that have no ready market and are held in perpetuity for their intrinsic yield, without a clear liquidation horizon unless they have a natural maturity through expiration or bringing to market. This means that the free cash flows from the intrinsic yield are the source of net income and therefore dividends to the owner, so any hedging must be funded against this cash flow stream, with the exception of collateralisation of the underlying asset, which even then can cause a distressed sale in case of negative mark to market of hedges. Although this is the most difficult area for hedging, it is clearly the most beneficial area as well, since the liquidation horizon stretches into several years. Any savings in risk horizon there, can totally transform the value of the business. We will review some of the characteristics required for successful hedging towards liquidation of private assets and likewise the fatal flaws where it simply won't work: namely Private Equity.

From a model perspective, we have a highly illiquid private asset, that we are valuing by a theoretical Discounted Cash Flow model with infrequent and possibly irregular cash flows that we can treat as a yield y which can be decomposed into market risk factors such as a discount rate, market spread, and then some intrinsic yield characteristics such as idiosyncratic risk and a liquidity premium. The yield may also be decomposed to additional risk factors such as inflation, default, and growth prospects. We then arrive at a total discounted value of π , where we look to collateralise the asset and receive funding $c_{\%}\pi$ as a liquid cash component, and acquire an offsetting hedge for the underlying risk factors with a continuous market value of h .

$$\pi - h = \sum_{t=0}^T \frac{y_t}{(1 + r_0 + s_0)^t} - y_0 \int_{t=0}^T (r_t + s_t - r_0 + s_0) e^{-r}$$

$$\Delta(\pi - h) = y_t + \Delta c_{\%}\pi - \Delta h$$

This means that the value of the hedge cannot be settled until the final sale of the asset, and therefore the interim value at risk is exorbitantly high, well exceeding the yield, even though the negative Value at Risk from the hedge implied a positive value increase in the asset.

Khan Paradox of Illiquidity Mismatch for Liquidity Value at Risk at Tau

$$f_t(r_{\pi_\tau}) = y_\tau + \Delta c_{\%}\pi - \Delta w_{t_0} Z_p \sigma_{h_\tau} \sqrt{\frac{T_{i_c}}{2}}$$

Thus, with Private Assets, we cannot hedge the value, but rather earn a Cash Flow yield, and then we hedge the discount factors of the cash flows which are mark-to-market, so the mismatch is yield – $\Delta \text{MTM}_{\text{hedge}}$. The hedge however is the summed value of those cash flows, since the model is effectively illiquid, it should be the Duration period of the liquidation, which again can be several years. Thus, at liquidation, it can neutralise the risk, but for any interim period, it will INCREASE the risk and create a funding illiquidity crisis equal to the value of the asset, even when the value of the asset theoretically has increased!

As a result, the difference between the two valuations, even when they are perfectly synchronized in terms of value and risk factors, will equal the infrequency of the cash flow multiplied by the change in the risk factor. Hence, we are ADDING risk, by adding a hedge, and creating a negative cash flow liability equal to the value of the asset, at a time when the value of the asset is increasing.

Let's take a quick example: a quarterly cash flow, and a risk sensitivity of 5 against a risk factor which will have a returns distribution that can lead to an expected loss of $-Z_p$, let's say -1 to be conservative. So, in a quarter, we can make $0.25y$ from the cash flow y over time increment Δt , while the hedge mismatch can grow to $5y$ at a time when the asset is increasing in value! This is notwithstanding the fact that it's a perfect hedge that would eventually resolve on the sale of the asset at an increased price. The interim mark-to-market of the hedge would create a loss.

In the next few sub-sections, we will look at the attributes that solve this paradox of interim value at risk, and then we will review how various asset classes respond to these and the appropriate measures for different asset classes. The various attributes that resolve this paradox are: steady high yield, low volatility, low idiosyncratic risk, high diversification and small sized instruments in the overall portfolio, ability to collateralise the underlying asset against the hedge, and the existence of actively traded long-term secured financing for the hedge instrument to last the course of the liquidation with a liquidity match against the sale.

High and Steady Intrinsic Yield for a “Natural Hedge”

The main problem with Private Assets is the lack of market-based transactions leading to a market valuation of the asset. Instead, all valuation methodology is accounting-based on a quarterly interval. Thus, we are not hedging the asset, but the income stream. This leaves a hedging mismatch against the private asset during the interim period that can lead to a liquidity crunch. If, on the other hand, there is a predictable fixed income stream from the asset, we can use that against the return function of the hedge and have a clear idea of how to match the fixed income stream against the hedge. In effect, we are creating a “natural hedge”, of the income from the asset against the hedge.

This is a common practice in Enterprise Risk, for businesses that deal with market risk factors, for example interest rate risk, currency risk, or commodity risk. We simply do the same for the predictable component of the income stream. This is necessary, but not sufficient: next, we would look to collateralise this income stream against the hedge, for the full duration of the hedge.

Low Volatility of the Market Risk Factor Compared to the Yield

The second problem with the matching of a market risk factor against a private asset, is how far the market risk can “run away” from the last valuation in the interim period. A highly volatile market risk factor, such as Equity, can move far too much to be a reasonable hedge. A low volatility of the risk factor does two things: it makes it easier to adjust against the yield and increases the ability to receive funding against the yield for the hedge. For example, rates, inflation, are likely to fit this category. If additionally the yield is much higher than the underlying risk factor, for example $y \gg r$, that also benefits the dynamics.

$$y_\tau + \Delta c_{\%}\pi > \Delta w_{t_0} Z_p \sigma_{h_\tau} \sqrt{\frac{T_{i_c}}{2}}$$

Where c is a cash component of the portfolio, or cash funded by collateralising the portfolio with a bank, effectively taking a loan against the illiquid asset, or parts of the portfolio that are acceptable for such a pledge.

Pledgeable Assets as Collateral against Self-Funding Hedge

As we saw from the previous section, the $\Delta c_{\%}\pi$ is the portion of cash or funding line received for collateralising the illiquid asset as a pledge in value against the hedge. Effectively, we are borrowing money against the illiquid asset, to fund the mark-to-market of the hedge. This funding line is more likely for assets that fit in a local bank loan portfolio, for example real estate or a loan portfolio that could be securitised in a similar manner to banking books. This also assumes that the assets are already unleveraged, so the loan-to-value means they are secured at the highest level for bank funding.

In a heterogeneous portfolio, the liquid assets may serve as pledgeable assets for the hedge, particularly in a fixed income fund, where highly liquid fixed income assets are ideal collateral for some more illiquid portions of the portfolio. The more illiquid, the higher the “haircut” of the collateral. We can expect the discount to be similar or more risk averse than the VaR liquidation function itself, so that the discount to value will be at least as high as we value it ourselves.

Finally, for single assets there can be embedded leverage at the asset level. Infrastructure, or large real estate projects, it makes sense to “drive down the hedge into the asset”. This means making the pledge to the hedge on the asset level. An example would be an infrastructure asset assuming a loan that pays for the interest rate swap, or a mortgage portfolio taking on a variable rate swap. Effectively, the collateral and hedging operation, is inside at the asset level, not at the portfolio level. This is quite common in infrastructure and real estate for interest rates and currency hedging.

Low Idiosyncratic Risk of the Portfolio compared to the Risk Factor Hedge

The fourth problem is in contrast with the first problem: we want $y \gg r$ but we still want a high correlation against the risk factor, so that we have a pure “alpha”, a constant premium in the yield. Statistically, that means we need to have a highly diversified portfolio of smaller components that comprise the yield, for example a diversified small-loan portfolio. Ideally, we wish yield to decompose to:

$$y = r + s = r + s_m + s_\varepsilon + s_\alpha$$

Where r is the market funding rate and s_m is the market spread, and both can be hedged against risk factors, leaving a spread “idiosyncratic” term $\varepsilon \sim \Phi(0, \sigma_\varepsilon^2)$ which limits to zero as the number of assets gets large $\lim_{n \rightarrow \infty} \frac{\varepsilon}{n} \rightarrow 0$, and α is a constant or at least time-varying positive “illiquidity premium” that is earned as an extra yield, found in the various assets comprising the asset class. So ideally, this should leave the yield paying for hedge, while still earning alpha. A good example would be a well-diversified collateralised loan portfolio. A bad example that does not satisfy this requirement would be a single company.

Minimum Correlation for Temporal Mismatch

The longer time obviously has a square-root effect against yield, but let’s also look at the correlation to understand the minimum correlation requirements for the hedge for a long-term hedge.

If we recall the correlation boundary for the time horizon:

$$\frac{T_L - \frac{1}{2}T_S}{T_L} < \rho$$

And then set the short-term horizon to the longest hedge horizon possible, under a standard CSA credit line with a Clearing Broker, and then set the expected correlation, we arrive at a maximum possible time horizon for an illiquid asset:

$$T_L < \frac{T_S}{2(1 - \rho)}$$

So, at perfect correlation, we can have an infinite time horizon into perpetuity. Anything less, and it quickly deteriorates, with 0.5 as the lowest acceptable correlation for equal length. A practical estimate for our longest short-term hedging horizon with a CSA Credit Support Annex at a Clearing Broker is 2 years, and then with an imperfect correlation of 0.8, we have a long-term liquidation horizon of 5 years. This is good enough for ALL of the aforementioned private assets. Thus, an appropriate correlation to regard as a minimum boundary, is 0.8.

Minimum Correlation for Full FX Hedge

The aforementioned temporal correlation minimum boundary is not required for the currency exposure, where we might wish to have a partial hedge during the liquidation. This does not relate exclusively to the liquidation but rather to the general risk management of the currency exposure of the portfolio. That said, the break-even correlation for a 100% hedge would be:

$$\kappa = \frac{\sigma_A}{\sigma_X}; h^* = 1 + \kappa \left(\rho_{A,X} + \frac{(r_d - r_f)}{\sigma_X} \right); \rho_{A,X} = -\frac{1}{\kappa} - \frac{(r_d - r_f)}{\sigma_X}$$

Liquidity Mismatch in Hedging Private Assets

The biggest operational risk to a private asset hedge is the liquidity mismatch between the market-to-market margin settlement of the hedge instrument on a regular basis, and the infrequent cash flows from the private asset, leading to insolvency. An infamous example is the liquidity gap caused by Metallgesellschaft's long energy futures positions leading to a billion dollar loss representing half the firm's equity, leading to insolvency and restructuring in 1993²⁷. This is likely to happen to enterprises active in international markets, usually from interest rate or currency hedges. The early design for the net sensitivity profile, prospective loss-making positions in hedges from the LVaR risk function, and finally the funding of hedges by the clearing broker via multiple broker-dealers during a liquidation, are crucial elements.

Khan LVaR Risk Function as a Liquidity Requirement

We have already established the VaR Risk function resulting from an unhedged linear liquidation. We can use this loss metric as the prerequisite cash buffer that is necessary as a liquidity component for the yield and cash collateral to cover. After all, the hedge is meant to cover this as an extreme move, hence it is as likely to make a profit of this amount as the risk function is to reach the loss; it's a symmetrical relationship.

Stressed VaR $\Phi(X < 1\%)$, $Z_p = -2.33$

Khan Liquidity Value at Risk

$$f_t(r_{\pi_T}) = \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{i_c}}{2}}$$

And then we set this to the weight of the liquid component w_l discounted to time zero:

Khan Liquidity Value at Risk LVaR Function for Illiquid Assets

$$y_t + \Delta c_{\%} \pi > e^{-r \sqrt{\frac{T_{i_c}}{2}}} \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{i_c}}{2}}$$

If we don't have this as a liquidity component, we need to be able to fund it from the portfolio as collateral for the entire course of the liquidation period, in this case several years. This will be used to fund the P/L account and mark-to-market margin of the hedge.

Long-Term Hedges: OTC Customer Support Annex

It's not just about having a hedge: it's about financing the hedge from the underlying asset. Exchange-traded securities don't allow this; we need to secure funding from a banking relationship in order to accomplish the hedge, preferably with multiple lines of credit from multiple broker-dealers to keep it honest. Short-term exchange-traded futures will surely be inappropriate, as was the case with Metallgesellschaft; any inability to settle margin will simply result in the hedge being liquidated. A long-term hedge will require a multi-year Credit Support Annex from a Clearing Broker that is funded by the asset itself, with appropriate collateralisation from the portfolio. In order to receive typical competitive terms in the CSA similar to market quoted prices for example for ISDA standard interest rate swaps and foreign exchange forward rates, a 2-year horizon is typical. For any longer periods, we must consider rolling forward rather than overpaying as is a natural inclination: thus, it should be expected that some income is being received during the liquidation horizon, where this can be applied against the hedge margin account.

Counterparty Risk of the Customer Support Annex and OTC Hedges

For a large banking entity, the CSA would be accompanied by a VaR function related to the risk of the non-settlement by the counterparty, as an XVA credit value adjustment. In our case, we would collateralise and assign it to a clearinghouse, so we choose to delimit this risk. The regulatory requirement is to have a number of counterparties for the OTC hedges in order to mitigate counterparty risk, with a maximum net exposure to any counterparty of 10% of NAV. For a bank, however, this is a major risk of conducting business, so for a larger player, the actual VaR amount is considered a counterparty risk, especially if the counterparty aggregates over a number of transactions, for example the top 10 counterparties would need specific treatment and insulation by means of credit default swaps against any default. A simple heuristic is to maintain a radar of the credit default swap level of every counterparty, and "red flag" the exposure once the counterparty reaches a certain level, for example 800bps in a stressed market. At that point, you would look to transfer the counterparty obligations to another bank, through the multiple broker-dealers attached to account via the main clearing broker. This was the case in 2008, when a number of smaller players gained ascendancy simply by not having been in the game before. Retail banks with underleveraged balance sheets would be examples, compared to the main broker-dealer banks, who might be overleveraged in stressed financial market conditions.

Private Equity: Fails at Every Level for Hedged Liquidation

The most intractable asset class for hedged liquidation is Private Equity. If we look at the various attributes required for a hedged liquidation, Private Equity as an asset class fails at every level. First, Private Equity does not have an intrinsic yield. Mostly, these are growth companies operating on negative cash flows, having a “burn rate” in order to achieve market dominance. In the cases where there is a stable cash flow, such as in LBO transactions and Mezzanine Private Equity, these acquisitions are usually already leveraged to the hilt for tax benefits, such that there is no room for any embedded leverage against any other financing activity. Second, the Equity Market risk factor has too high underlying volatility in the returns of both the hedge and the asset class. The VaR function for the Equity Market is such that it allows for a loss of 35% in a single quarter, which means it's not possible to have a long-term liquidation plan. Third, it is unlikely that a single corporate asset will serve as a reasonable pledge for bank funding of a hedge towards disposal, mainly due to the overleveraged nature of the collateral already; it does not meet the loan-to-value criteria for collateral. For the same reasons, it is unlikely that the hedge can be “pushed down into the asset”, due to the overleveraged nature, and also because the risk factor itself does not correlate well with the business. Fourth, the type of hedge instruments used in equity are mainly short-term exchange-traded futures, which need to be settled on a day-to-day basis, and not likely to be of interest for a bank to take on the counterparty risk in the course of a liquidation. From the temporal correlation boundary condition, the future is 0.25, while the liquidation period is 5, means that the correlation must be 0.975, which is never the case in Equity Market Beta; it's more likely to be 0.5 in correlation, less than what it is for exchange-traded equity. Which leads us to the last and final point, the idiosyncratic risk is as big as the market risk and unlikely to be diversifiable. The size and concentration of private equity investments means that there are no simple ways to diversify the idiosyncratic risk of what remains a handful of super-sized investments, often in the same sector, overleveraged, or in need of distressed restructuring.

For this reason, Private Equity tend to be “one-way street” Closed-End funds, where there are no redemptions, but rather a lifecycle of investment and divestment for a 5-10 year term which constitutes the life of the fund. The few times where Private Equity has been mixed in with a heterogeneous Equity fund, have been an unmitigated disaster, for example the Woodford Equity Fund²³, in the UK. In that case, micro-caps with a nominal listing but no underlying trading, and hence were essentially private equity (without even the benefit of a capital injection from a large private equity firm), were part of a larger retail fund, and the subsequent redemptions left a large unredeemable private equity component. This is ill-advised. In the case of a fiduciary asset manager overseeing ownership on behalf of pension funds for example, the appropriate action would be an in-specie transfer of the ownership of the corporate entities in the event of a redemption or asset transfer. The best that can be offered is a side-pocket and an attempt to find a buyer in the secondary market for each company, which generally takes years.

Private Credit: Passes at Every Level for Hedged Liquidation

The most amenable asset class for hedged liquidation is Private Credit, especially when it is pooled into well-funded unleveraged structures that have look-through to the underlying loans. It is one step away from being a CLO or an MBS. In a sense, it is waiting to be “taken to market”. Private Credit encompasses semi-liquid exchange-traded fixed income structured credit securities, mortgage-backed securities, collateralised senior secured direct lending, collateralised trade and secured merchant finance, insurance-linked mezzanine finance, auto loans, retail consumer finance. As such, Private Credit is basically the banking book, carried out by non-bank participants, in other words “shadow banks”: lacking the transparency and regulation of banks, but also lacking the leverage. These are essentially over-collateralised, under-leveraged loans funded by equity from asset owners. This implies that the investment portfolio lends itself well to acquisition and financing by banks. The main problem with Private Credit is periods of systemic stress when there will be no appetite for risk transfer, hence making it a pro-cyclical risk. But let’s look at the benefits first and then approach the challenges.

First, Private Credit has an intrinsic yield and a short-term fixed maturity, that can be viewed as an inter-bank funding rate, a credit spread, and an illiquidity premium. We can state it in the same way as earlier Public Credit models, and then simply diversify the small idiosyncratic effect by creating a larger pool of these, so a summation of many. The lending contract is stated exactly in these terms, as a funding rate plus 500 bps. The yield risk factors are already partially hedged with the variable funding rate to the clients, which leaves credit and default risk as the main risk factor. Lending is also often secured against the business at a senior level for collateral against the business and property to reduce the expected loss at a prospective default. When placed into a fund, and listed on an exchange, the shares of the fund are exchange-traded structured credit CLOs.

We have the same yield model, weighted and summed over many component obligors:

$$\sum_{w_i=0}^1 y_i = \sum_{w_i=0}^1 w_i (r + s_i = r + s_m + s_{\varepsilon_i} + s_{\alpha})$$

And the same risk factor mismatch that serves as a constraint for the liquidity, where we need not just the yield but a cash component to manage a loss due to liquidity constraints:

Khan LVaR Risk Function for Private Credit

$$y_t + \Delta c_{\%} \pi > e^{-r \sqrt{\frac{T_{lc}}{2}}} \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{lc}}{2}}$$

Second, we have a low volatility in the risk factor, other than the jump risk due to illiquidity. The actual rates factor is the most low volatility risk factor of all: it is Duration and Leverage that create a risk out of this market factor, both of which are possible in the first place because of the low volatility. If we can control or adapt Duration and Leverage, we can manage the aftermath.

Third, although the private assets do not lend themselves well to pledging against collateral, they are in the same Fixed Income family as higher liquidity securities which therefore fit well in an overall Fixed Income portfolio. It would not be a far reach to mandate a portion of a Private Credit portfolio to be “Dry Powder” in the form of marketable, highly liquid fixed income securities such as Financial Bank instruments, since they effectively are the same underlying asset in terms of look-through. With such a component, you could then pledge the liquid portion as collateral for the hedging of the Private Credit portion as well and expect the same risk factors at play in both segments. Again, it would demonstrate “Right-Way Risk” because both the more liquid pledged collateral and the illiquid Private Credit would behave against the same market risk common factors that would be perfectly negatively correlated with the hedge.

Fourth, in terms of idiosyncratic risk, the Private Credit portfolio is made of granular small sized credit commitments that are pooled in the thousands to create a well diversified risk profile, analogous to indexation in equity markets. If we look at the earlier model for the risk, the idiosyncratic risk will tend to cancel out, leaving systematic risk factors, which we wish to respond to in terms of the hedging instruments.

Fifth, looking at the correlation between the hedging instruments and the credit portfolio, the actual valuations models for both will be similar, and therefore they should correlate well against each other, meeting the temporal requirement boundaries for a long-term hedge. The main difference is the liquidity mismatch and valuation mismatch, such that it will only work in a long-term hedge, theoretically until maturity if there is no interim liquidity. So, the hedging instrument is constrained more by the MTM mismatch in case of funding calls, than the correlation. This is not quite accurate for highly idiosyncratic, specialised credit, so we caveat this discussion to highly identifiable credit spreads and explore other types of idiosyncratic separately as other forms of private assets.

Discounted Cash Flow Valuations of Illiquid Idiosyncratic Assets

We will move the discussion to pure illiquid assets with no market value equivalent and highly unique idiosyncratic properties, whereby the only valuation is a irregular Discount Cash Flow. Even then, we can see the similarities between a DCF valuation and a bond. They are both the same valuation, except the illiquid asset doesn't have a clear cash flow component so we are predicting it, and the actual market equivalent yield is also a best estimate.

Real Estate: Potential Short Credit

In the earlier section, we spoke about equity-funded Real Estate as “Potential Credit”, and a moderate increase in leverage could be seen a short Credit position. If we look at the DCF Model we are using, it can value a bond, a piece of real estate, or a utility. Thus, real estate might be an equity asset, but it lends itself well to a fixed income valuation. What differs is the predictability of the yield, and the idiosyncratic component of the spread risk factor, which increases with the uniqueness of the fixed private asset.

$$\pi = \sum_{t=0}^T \frac{y_t}{(1 + r_0 + s_0)^t}$$

So, for real estate, we can estimate the rental income as a capped lagged inflation-hedged cash flow, and the spread as a small entity high risk credit spread. So, we can still price it as a credit instrument, but with a larger idiosyncratic component. If we look at how this approach scales to a larger real estate portfolio, we have characteristics of Yield, Vacancy, Weighted Average Unexpired Lease Term, any covenants on Inflation Adjustment to Lease or Rent, and then most idiosyncratic of all, the Default Risk of the tenant.

Look-Through to Tenant Quality as Credit Quality

One of the big risks in managing a real estate portfolio is the quality of the tenant in terms of credit quality. The idiosyncratic risk can therefore be modelled as actually the tenant being the ultimate obligor risk. This can be modelled similar to a credit rating, in terms of income coverage of the rental liability and counterparty risk on the aggregate level in terms of an investment mandate. This is an approach commonly used by Banks and Pension Funds, whereby a look-through on the properties will reveal credit quality and counterparty concentration in the leaseholders across a property portfolio.

Geographic and Sector Concentration in Real Estate

A secondary concern is geographic concentration and sector concentration across a property portfolio, again perceived from the perspective of reducing idiosyncratic risk. Investment mandate restrictions on the sector and geography can enforce a diversification that mitigates any idiosyncratic risk.

Loan to Value Metric

The most important metric in terms of “Potential Credit” is the current Loan to Value of the real estate assets. This can be a guiding principle in the acquisition of the real estate assets in the first place, enabling a more efficient liquidation over a year-long liquidation horizon. The leverage becomes cash and a short credit position:

Khan LVaR Risk Function for Real Estate

$$y_t + \Delta c_{\%} \pi > e^{-r \sqrt{\frac{T_{ic}}{2}}} \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{ic}}{2}}$$

Infrastructure: Private Sovereign Specialised Revenue Bond

With utility assets, we have an equity holding in a unique and therefore highly idiosyncratic asset with a perpetual life, but equally we have predictable and often regulated revenues due to local monopoly effects. This means we have a stream of predictable cash flows, but with more idiosyncratic risk factors related to the underlying utility provision. What is important, in comparison to real estate risk factors, is the lack of default risk other than from the self-assumed leverage of the enterprise. Again, a utility lends itself well to a Fixed Income type of valuation and leads to hedging the common risk factors. Infrastructure is essentially a private sovereign specialised revenue bond.

If we look at the same DCF model that we used for Real Estate and for Bonds:

$$\pi = \sum_{t=0}^T \frac{y_t}{(1 + r_0 + s_0)^t}$$

We again have a lagged inflation-linked regulated yield, and some risk factors, that do not involve default. The risk factors would be the underlying funding rate, the credit spread for a similar fixed asset, and the idiosyncratic costs of provision of the particular utility, which may link to energy costs, macroeconomic conditions, even socioeconomic population decline or sociological trends like environmental preferences.

Leverage as Reverse Credit: “Driving Down Hedges into the Asset”

In the case of a unique infrastructure holding, we cannot securitise it easily, but we can drive down hedges into the asset. In the case of the long asset position, we can release some equity as a cash buffer, and hence create a short credit position, whereby the long duration position of the asset is cancelled by the short duration created from the financing, hence making the asset duration neutral during the liquidation. Leverage becomes cash and the short credit position:

Khan LVaR Risk Function for Illiquid Fixed Assets

$$y_t + \Delta c_{\%} \pi > e^{-r \sqrt{\frac{T_{ic}}{2}}} \Delta w_{t_0} Z_p \sigma_{\pi T} \sqrt{\frac{T_{ic}}{2}}$$

FX Hedge in the Infrastructure Asset

If we are in a position to be a strategic foreign investor in an infrastructure asset, we can drive down a currency swap into the asset itself, rather than hold it separately. We can take funding against the enterprise in terms of leverage, to create a cash buffer which funds the FX hedge.

Khan FX Hedge Formula (2015)

$$\kappa = \frac{\sigma_A}{\sigma_X}; h^* = 1 + \kappa \left(\rho_{A,X} + \frac{(r_d - r_f)}{\sigma_X} \right)$$

We can make this a company-level liability, driving the hedge into the enterprise cash flows, and then releasing a cash buffer with a moderate leverage against the revenue stream, thus mitigating risk factors of currency and interest rates, over a multi-year liquidation horizon, potentially into perpetuity as an optimal financial structure holding for this type of asset.

Measuring Idiosyncratic Risk with Default Rate

As we have strived to create Private Assets Portfolios of numerous granular holdings that amount to a small portion of the overall investment profile, we might come under the impression that we have eliminated idiosyncratic risk. A better approach, is to assume some kind of innate imbalance in the portfolio, driven by how a continuous default rate λ will affect both returns and the basis risk factor. We can use the default rate as a loss rate on the portfolio as well as a cross-sectional deviation on the returns $\sigma_\varepsilon = \lambda$, such that there is always an error term measured in the basis risk, as opposed to a jump risk. This would then express itself at a long-term time interval frequency equal to $1/\lambda$. So, for example for a default rate of 1%, we would keep a cross-sectional deviation of 1%, and also a return drag of 1% in expected return. We can use the formula as was detailed in the Khan Interest Rate Calculus⁵.

The Loss Given Default is not the full Notional 1, but rather $1-R$ (Recovery Rate), which is decided by Collateral and Loan-to-Value metrics, which usually result in a Recovery of 50%. Hence, the spread is a compensation for the annualised probability of default such that the spread is earned up successfully over time until it has a chance of zero at longer time due to exponential decay, with the decay rate:

$$\lambda_\tau = \frac{S_T}{1-R}$$

This gives us a total probability of default over time T :

$$\prod_{t=0}^T Pr(default) = 1 - e^{-\lambda T}$$

We can infer the market-traded spread to be an indicator of the implied probability of default:

Khan Formula for Spread Implied Default

$$s = \ln \left(1 - \prod_{t=0}^T Pr(default) \frac{R-1}{T} \right)$$

And thereby choose a point, for example a Probability of 50%, to be the pivot from the Spread earning a Carry, to being a liability that earns a Lamda that cancels out the Spread. Thus, at a Spread above an implied probability of default of 50%, the Spread stops earning a Carry in the Performance Attribution and P&L Explain.

A larger question is if we should weight all spread return by the probability of default, but that appears not to be the case empirically. Rather, some of the bonds turn out to be subject to default risk, not all. And we catch those with high λ once the spread convinces us of the probability of default dominating the price.

Limits to Hedging: Leverage and Incomplete Markets

There are three main limits to using hedging mechanisms for liquidating illiquid positions. First, there are regulatory limits to using hedging since hedges are measured as leverage. For asset managers, there is a tendency to legislate “long only” fund vehicles, that are not allowed to carry short positions, even when those reduce risk. Second, in those cases where funds are allowed to have hedges, they use it as leverage to increase their exposure! As seen in earlier sections, adding hedges are a good way to isolate the unique properties of an investment, what we call idiosyncratic risk is often treated as “alpha” by speculative funds, as though it is a time-invariant constant. And finally, perfect hedges might not exist, as we saw in the hybrid approach to illiquid fixed assets, what in theoretical finance is called incomplete markets.

Regulation on Leverage

There is a current lack of sophistication in long-only asset management, insurance companies, and the attendant regulatory framework to enable the use of the hedging techniques expounded in this research. Most regulatory frameworks have limits on gross exposures and therefore leverage, even when those gross exposures are negative and therefore netting the overall exposure. Examples include US mutual fund regulations imposing a maximum leverage, and EU retail fund regulations imposing a maximum leverage as well, both of which make it regulatorily difficult to implement the techniques recommended here. There is a further imposition on liquidity timelines, which ironically mean that a forced sale is incentivised over a smooth hedged liquidation.

The solution is two-fold. First, for asset managers and insurance companies, we can update the regulatory framework to impose limits on net exposures versus gross exposures. This is a trend already underway, for example with the Net Exposures and Sensitivity measures being promulgated and adopted. If these were measured by a third-party, such as a custodian bank using an independent software, that would assist in the veracity and independent accountability of such analytics. Second, banking participants who are not constrained by the same regulatory framework can provide the service for such non-financial participants, via market-making and custodian help, kind of like an expanded Prime Services for these investors. This would involve collateralising the portfolio and holding it in the interim of a liquidation. This activity could be promoted by Prudential Regulation in order to assist in smooth running of financial markets.

Speculative Nature of Active Long-Short Funds

For speculative active long-short funds, we can expect that they are already implementing hedges, in order to maximise the spread trade, to the extent where they are fully leveraged to the maximum potential. This means that if they miss a margin call, and they are forced to liquidate, then there is no way to implement further hedges for the liquidation, thus forcing a fire sale. A good example of this is Long Term Capital Management with sensible long-short spread trades, and a reckless amount of leverage²⁸: to make 20% on a 20bp swap spread, a speculative fund is taking 100x leverage.

A standing joke among hedge funds, is that everyone's question is "Where's the Hedge?". This is because the hedge is used to isolate the risk factor and then take leverage to increase the exposure to the idiosyncratic risk. We have seen how to reduce the volatility from σ_π to σ_ε , but what speculative funds do is increase the leverage to $\frac{\sigma_\pi}{\sigma_\varepsilon}$. Thus, for example an 80% reduction in risk, would be followed by a 5x increase in the leverage, to maintain the original asset volatility. Furthermore, this leverage would be done on historical unconditional estimates, such that a slight spike in the asset volatility would lead to inability to meet margin calls and consequently a fire sale.

The solution is two-fold. First, to have a prime bank and depositary bank in charge of the asset, whereby the fund or investor company pledges the asset. This will ensure that the assets are unencumbered by other debt such that the hedge is for one purpose only, namely to reduce risk, not to increase it. Second, to ensure that the hedge is in fact for liquidation, not to increase the risk in the underlying spread factor. This helps in two ways. First, in the case of actual redemptions in a fund, the sale of the assets could be side-pocketed together with the hedge, and then left to the ultimate investor in the case of the bank market-maker. Second, it makes the bank and active participant in the underlying market, which creates more liquidity and helps in making complete markets.

Incomplete Markets

In our liquidity analysis, we wish to separate "slow markets" from "no markets". In other words, the idiosyncratic spread occur not because of low liquidity, but because of no liquidity. There are certain assets missing, from the hedging universe, from the risk factors, that means that we are unable to hedge out the risk with other securities or instruments. The objective of complete markets, is the introduction of liquidity first from more market participants, but more importantly from actual creation of securities design to mitigate risk in exactly the type of holdings that an investor would invest in. Think of it as an island, which has only a private airfield. We are looking to create more of a hub, for everyone. This will inevitably reduce the idiosyncratic spread that makes the asset appealing, but it won't disappear, it will simply be made available through mutualisation to a larger group of investors. As we've seen in private credit, it might very well disintermediate the banking debt and supplant these securities with higher yielding credit for a wider range of investors.

Summary on Hedging for Liquidation

We see that the bond calculus can still make use of the same measures developed earlier but separated into a rate component and a spread component. Thus, a portfolio can be valued by the aggregation of a single measure, as described by the Khan Par Yield Duration Formula (Khan 2016⁵):

$$\text{Khan Par YTM Duration } D_{yT} : dP_{\%T} = \frac{e^{-yT} - 1}{yT} dy_T$$

But applied differently for changes in rate, spread, inflation, or callability. One caveat regarding credit default, is the Loss Given Default, which is added at a probability threshold.

$$\text{Khan Key Rate Spread Duration} : dP_{\%T} = \sum_{t=0}^T \Delta D_{\Delta t} ds_{\Delta t}$$

Which gives us a Khan P&L Explain Formula as follows:

$$d\pi_{\%T} = \sum_{t=0}^T \Delta D_{\Delta t} dr_{\Delta t} + \sum_{t=0}^T \Delta D_{\Delta t} ds_{\Delta t} + \left((s) - \ln \left(1 - \prod_{t=0}^T Pr(def > 50\%) \frac{R-1}{T} \right) \right) dt + ds_T + r_\tau dt + dr_\tau$$

Where expected spread return is adjusted for probability of default.

We then take a hedge on the market rate factor and the market spread factor:

$$\text{Short Market Rate and Spread} = -dh_{\%T} = -\sum_{t=0}^T \Delta D_{\Delta t} dr_{\Delta t} - \frac{e^{-rT} - 1}{s_h} ds_T - (s_\tau) dt - ds_\tau$$

So, we have $d\pi_{\%T} - dh_{\%T} = \left((s) - \ln \left(1 - \prod_{t=0}^T Pr(def > 50\%) \frac{R-1}{T} \right) \right) dt + r_\tau dt + dr_\tau$

Which leaves a residual risk of the idiosyncratic term, which we can express in two ways:

$$\text{Correlation-based: } h^* \sigma_h = -w_\pi \sigma_\pi : \sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

$$\text{Duration-Neutral Idiosyncratic Spread: } h^* \frac{D_{s_\tau} \sigma_{s_h}}{D_{s_h} \sigma_{s_\tau}} = -w_\pi \rho$$

$$y - h = w_\pi (r + s_i) - w_h (r + s_m) = w_\pi (r + s_m + s_\varepsilon + s_\alpha) - w_h (r + s_m) = w_\pi (s_\varepsilon + s_\alpha)$$

$$\sigma_\varepsilon (y - h) = w_\pi dP_{\%T} = w_\pi \frac{e^{-(r_T+s)T} - 1}{(r_T + s)} ds_\alpha$$

Liquidity Risk Reporting and Stress Testing

We can now summarise the techniques developed into a Liquidity Risk Report for Investing and Regulatory risk reporting and stress testing. A good example is ESMA's Liquidity Stress Testing Guidelines for UCITS and AIF funds²⁴. To satisfy this requirement, we use the Khan Waterfall methodology with exponential unconstrained unwind liquidation, as a stressed liquidation measure. We will also add the Hedging Measures in a hedged linear constrained unwind to compare and distinguish the potential prospects available as the industry mechanisms and regulatory risk requirements evolve. We will use the calculus summarised in the previous page, to calculate the rate volatility in terms of annualised P/L, the hedge amounts, and the stressed liquidation Waterfall and Hedged LVar99 including all Transaction Costs.

Let us demonstrate the techniques with a Global Rates Fund containing Credit elements as an all-encompassing universal example, with a \$20B liquidation to cause liquidity issues in even the most liquid instruments. The hedges are performed with OTC Swaps to neutralise the Rates sensitivity in terms of Duration but even Bond Futures in the most liquid markets is a good choice, and OTC CDS Index Swaps to neutralise the DTS (Duration times Spread) sensitivity of Credit Spreads (for the credit holdings, which will need two separate hedges: one for the underlying rate and another for the credit spread).

Methodology

Risk Exposures

We measure a portfolio in terms of notional percentage weights, $\sum_{i=1}^n w_i = 100\%$, grouped into 5 Liquidity Groups from most liquid to semi-illiquid: Government Bonds, Corporate Investment Grade, Corporate High Yield, Mortgage-Backed Securities, Asset-Backed Securities, and Structured Credit (CLOs, Securitised Private Credit). The credit and liquidity grade is classified by the ESMA Basel Credit Quality Step, CQS, from 1 to 4, as an indication.

Risk Hedge Factors and Stressed Market Shock

We use the summarised formulas from the previous section to create a P/L% for an annualised kurtosis adjusted volatility K-Vol¹², which is then multiplied by the weight to create a Risk Weighted Exposure. We use the par yield for each instrument to create a Rate Duration, Credit Spread Duration, and a residual idiosyncratic spread, weighted for each liquidity group. We then create a Weighted Duration times Spread Sigma by taking the residual idiosyncratic spread multiplied by the K-Vol to obtain a residual P/L separate from the market risk factors of Rate and Credit Spread. Next, we use a Stressed VaR $\Phi(X < 1\%)$, $Z_p = -2.33$ for the stressed market shocks during liquidation, for a Market Impact stress for the Rate and Credit risk factors as well as the Idiosyncratic Spread risk factor which is listed as the Hedged residual exposure.

Transaction Costs

Next, we compile transaction costs from TRACE²⁹ data in bps for each instrument by Liquidity Group. The hedges are OTC Interest Rate Swaps and CDS Index Swaps, both of which are cleared to eliminate counterparty risk, and funded by collateralising the liquidation portfolio. The hedge for the Credit Instruments requires two hedges, one for the base rate component and another for the credit component, unless the base rate is a variable rate component for the credit instrument, as it is for CLOs. The stressed amount is taken to be 4x the normal market bid-ask spread, based on market dynamics during historical stressed periods taken as indicative.

Liquidation Horizons

Next, we take the stressed liquidation horizons in days for each of the Liquidity Groupings, for a total portfolio liquidation of \$20B. We take an estimate of the daily volume from TRACE data and plan a liquidation schedule in accordance with a Waterfall exponential unwind methodology and a Hedged linear unwind methodology. Although this liquidation is large enough to cause liquidity issues, JP Morgan's CIO Office "Beached London Whale" had more than \$150B in exposures³⁰ for liquidation, so stressed liquidation scenarios can be much larger, and likely require longer liquidation horizons following the formulas set in the earlier sections. The trading horizons here are suitable for liquid and semi-liquid unwinds of up to \$50B with these portfolio constituents.

Stressed Liquidity Value at Risk LVaR99%

Finally, we combine the time adjusted stressed value of the market shock with the transaction cost to arrive at a Hedged LVaR99% and an outright LVaR99%. We separate the market shock from the transaction cost for a loss attribution in order to identify the loss drivers and compare the hedged risk profile with the outright risk profile.

Liquidity Dashboard and Charts

The final deliverable for the liquidity dashboard is the summary table and charts. We present the results in visual insights for the risk reduction in Hedged LVaR99% versus outright LVaR99%, with an idea for how much each liquidity group contributes to the loss attribution. If we walk through the charts, we start with a Weights Exposure by Bond Type, compared with a Risk Weighted LVaR by Bond Type, as well as a Hedged LVaR by Bond Type. We then look at a comparison of the loss amounts of the Waterfall LVaR compared to the Hedged LVaR by Bond Type. For the outright Waterfall LVaR, we also look at the Market Impact versus the Transaction Cost by Bond Type. Finally, we look at a Loss Attribution chart for the LVaR Contribution of each Bond Type, for both the Waterfall LVaR and the Hedged LVaR.

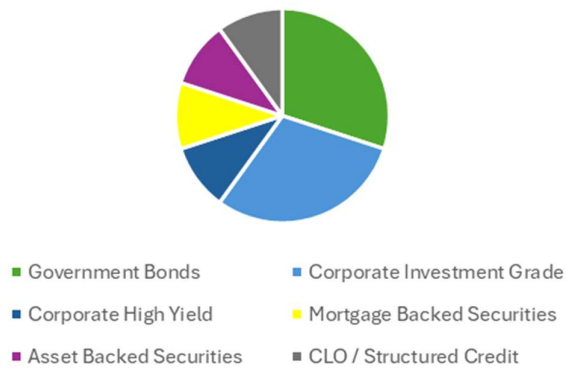
Global Rates Liquidity Risk Report

Risk Exposures and Hedge Factors by Bond Type

RISK STATISTICS				Credit	YIELD	RISK HEDGE FACTORS			
Bond Type	EXPOSURE	RISK (%P/L KVOL)	Risk Weight	Quality	Yield%	Duration	Credit	diosyncratic	wDTSS
Government Bonds	30%	5%	1.50%	1	4.5%	-8.1	0	0.30%	0.02%
Corporate Investment Grade	30%	9%	2.70%	2	6.5%	-4.3	-4.3	0.50%	0.05%
Corporate High Yield	10%	11%	1.10%	3	7.5%	-4.2	-4.2	1.50%	0.17%
Mortgage Backed Securities	10%	7%	0.70%	3	7.0%	-10.8	-10.8	0.75%	0.05%
Asset Backed Securities	10%	9%	0.90%	4	8.0%	-4.1	-4.1	3.00%	0.27%
CLO / Structured Credit	10%	11%	1.10%	4	9.0%	-4.0	-4.0	3.00%	0.33%
TOTAL	100%	8%	8.00%		6.5%	-6.0	-3.6	1.07%	0.10%

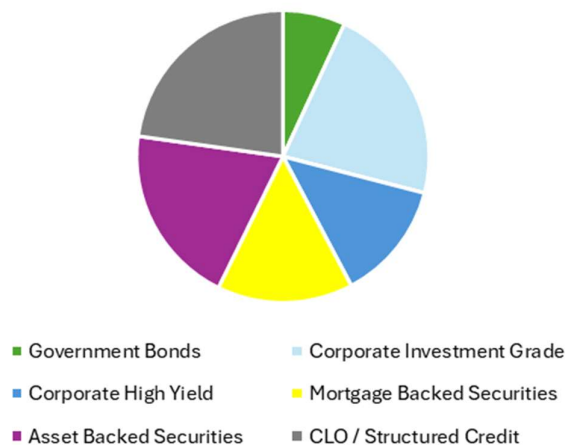
Exposure Chart by Bond Type (Weight in %)

EXPOSURE by Bond Type (%)



LVaR99% by Bond Type (% Contribution)

LVaR by Bond Type (% Contribution)

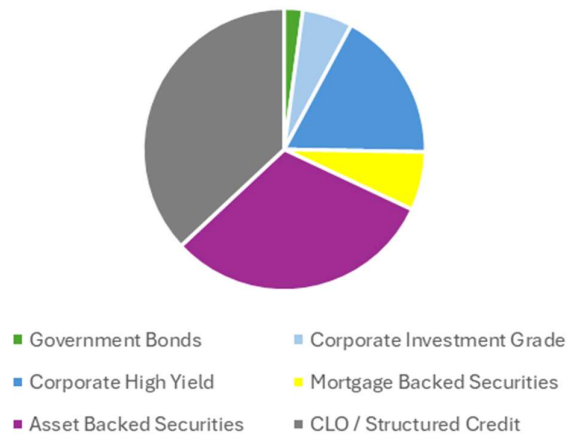


From an exposure of 2/3 in highly liquid investment grade securities, we see that the actual risk from a liquidation perspective is in the 3 semi-liquid groupings. The LVaR for these 3 groups

amount to 2/3s of the portfolio risk, even though they hardly amount to 1/3 of the weighted exposure.

Hedged LVaR99% by Bond Type (% Contribution)

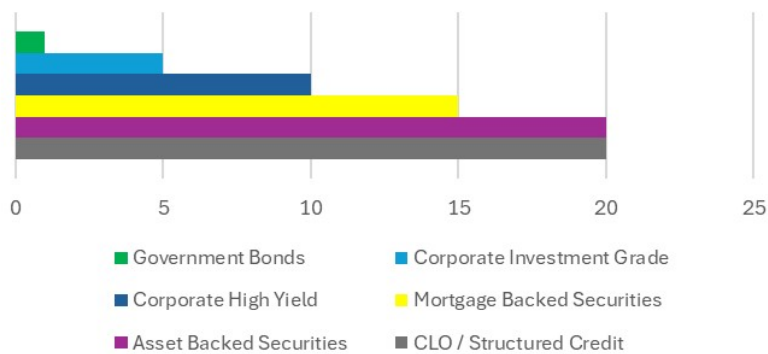
Hedged LVaR by Bond Type (% Contribution)



For the Hedged LVaR, we observe that the risk is even more concentrated in the illiquid portions of the portfolio, since these represent the residual idiosyncratic risk factors. But the actual residual risk that remains once the market risk factors are hedged and the liquidation is hibernated over a long-term liquidation is a fraction of the overall risk.

Liquidity Horizon Chart

Liquidity Horizons
(Days to Liquidate)

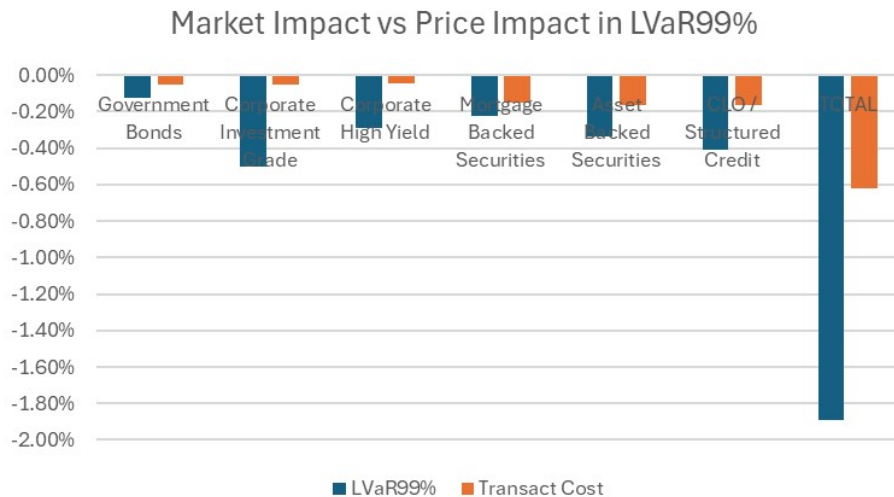


The heterogeneity in time to liquidation is highly varied in Global Rates, from 1-2 days in the highly liquid, one to two weeks in the medium corporate issues, to a month or more for the semi-liquid holdings.

Stressed Market Shock, Transactions, and LVaR99% Table

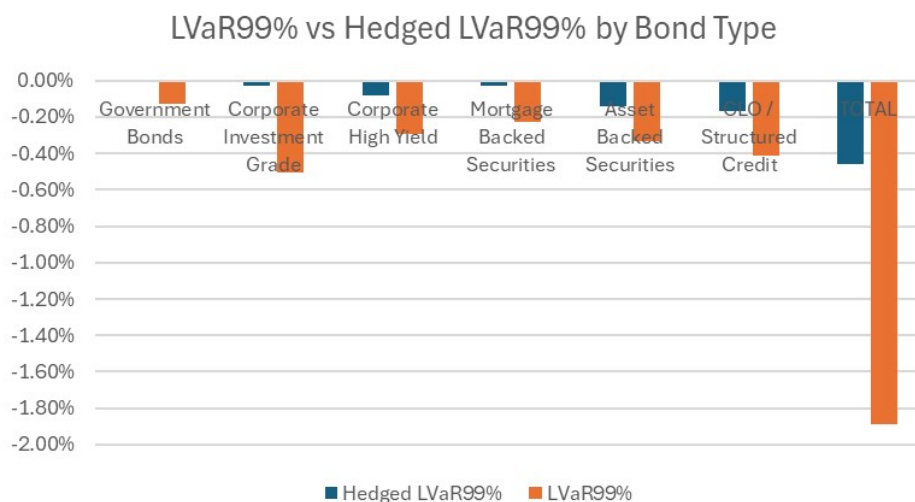
Bond Type	STRESS (Yearly)		TRANSACTIONS Normal (Stress 4x)		LIQUIDITY		Stressed Liquidity Value at Risk LVaR99%			
	LVaR99%	Hedge LVaR99%	Bid Ask bps	P/L %	Hedge	Time (Days)	Hedged LVaR99%	LVaR99%	Transact Cost	LVaR99%wTC
Government Bonds	-3.50%	-0.03%	1	-0.04%	2	1	-0.01%	-0.13%	-0.05%	-0.17%
Corporate Investment Grade	-6.29%	-0.10%	2	-0.04%	7	5	-0.03%	-0.50%	-0.05%	-0.55%
Corporate High Yield	-2.56%	-0.38%	5	-0.10%	7	10	-0.08%	-0.29%	-0.04%	-0.33%
Mortgage Backed Securities	-1.63%	-0.12%	7	-0.38%	7	15	-0.03%	-0.23%	-0.15%	-0.38%
Asset Backed Securities	-2.10%	-0.63%	20	-0.41%	7	20	-0.14%	-0.34%	-0.16%	-0.50%
CLO / Structured Credit	-2.56%	-0.77%	20	-0.40%	7	20	-0.17%	-0.41%	-0.16%	-0.57%
TOTAL	-18.64%	-0.10%	6	-0.15%	6	8	-0.46%	-1.89%	-0.62%	-2.51%

Market Impact versus Price Impact for Outright LVaR99%

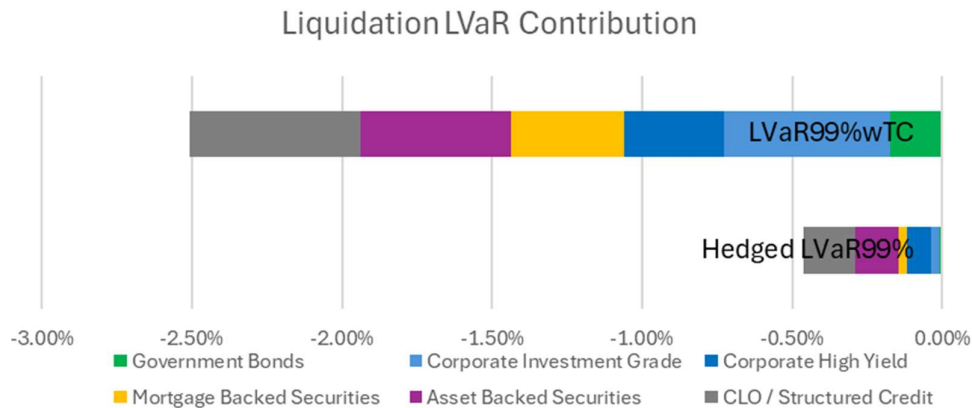


An interesting insight from the breakdown of market impact versus price impact for the accelerated Waterfall outright liquidation, is that even with the fastest possible liquidation, the market impact is 4 times larger. Therefore, any reduction in the market impact by hedging risk factors, will dominate the decision-making process.

LVaR99% versus Hedged LVaR99% by Bond Type



Liquidation LVaR Contribution for Outright LVaR99% and Hedged LVaR99%



From the last two charts, we see that a hedged liquidation dominates an outright liquidation by reducing the LVaR loss potential by 80%. Even though the components are equally balanced between the bond types, the actual amount is much less by hedging out the market risk factors for funding rates and corporate credit. Adding the insight from the last chart, we see that the overall amount is even less than the transaction cost for the Waterfall outright liquidation, meaning that even with an instant liquidation, the Hedged LVaR is lower. This happens because the transaction costs for the hedging instruments are less than for the portfolio assets.

From this analysis, we can ascertain that if we have an ability to hibernate a liquidation by allocating a liquidation against a market-neutral hedge with a clearing broker, it would be an expedient option to reduce the risk and loss potential in terms of stressed LVaR. In the case of an investor redeeming from a semi-liquid fund, it would involve that the waiting time would be longer, but the loss potential would be 80% less. It might actually give more confidence to investors that are trying to have first-mover advantage, and create more patience and less panic when it comes to fund holdings. As such, it would make for a prudential safeguard in fund redemptions as a “hibernation” technique that can be added to the prudential framework.

Khan Theorem of Economic Efficiency in Market Impact

There is an implied theorem of economic efficiency related to the trade-off between the market uncertainty and the trading costs, and the ability to hibernate a liquidation with a market-neutral hedge complement. If we can balance the trade, then so can a specialised trading market-maker bank counterparty avail themselves even more of the market tools we develop, simply because of their match-maker capacity of bringing together more players. Thus, in an ideal world, the ability to hibernate a liquidation with a hedge complement, should translate into lower transaction costs for the illiquid individual bond itself. The gap will be narrowed by technology, as open banking standards, automated processing protocols of standardised data attributes in financial contracts, and fintech channelling for funding syndications and rates marketplaces will enable bank to have a wholesale market for this type of funding. Likewise, there would be an incentive from a prudential regulatory perspective to allow market participants to provide this service in liquidation in order to encourage and maintain smooth and liquid financial markets within a banking system.

We avoid developing a general equilibrium model; what we do instead is formulate the economic incentives for a transaction cost in terms of hedging efficiency, risk reduction, and the resulting “Good Deal Boundary” that emanates from a hedging portfolio. Thus, a market-maker could acquire a position, slap on an offsetting hedge to hibernate the exposure and hold the position indefinitely to make a profit from the transaction by carrying the position through its natural liquidation time horizon. This would bring overall transaction costs down, for all assets that have suitable hedging counterparts.

Khan Good Deal Heuristic for Transaction Cost in a Hedged Liquidation

$$TransactionCost_{Hedged\%} \sim 4 \cdot KVaR(1-p) = \Phi(X < p) = -Z_p \sigma_\varepsilon, \text{ where } \sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

Which includes the trade in the outright asset, as well as the hedge. Hence reducing:

$$TransactionCost_{Outright\%} \sim KVaR(1-p) = \Phi(X < p) = -Z_p \sigma_\kappa, \text{ where } \sigma_\kappa = \sqrt{\frac{\kappa_*}{3}} \sigma_*$$

So that a market-maker should be able to take on the Outright with a Hedge Basis Risk:

Khan Theorem for Hedged Basis Pricing Outright Transactions

$$TransactionCost_{Outright\%} \sim 4 \cdot KVaR(1-p) = \Phi(X < p) = -Z_p \sigma_\varepsilon, \text{ where } \sigma_\varepsilon = w_\pi \sigma_\pi \sqrt{1 - \rho_{\pi,h}^2}$$

Implying that the transaction cost for the outright would be the holding period for minimising the transaction cost in the outright not by its underlying market risk but rather the hedged basis risk, so that a market-maker can have the patience to hold on to the market-neutral exposure and liquidate it during a longer more patient holding period. This is evidently not the case in current market dynamics but given the market mechanism delineated and the results in the liquidity risk report for the two liquidation approaches, they should converge. Prudential regulatory incentives can also provide directives and allowances towards this outcome, whereby a depositary bank would be allowed to hibernate a liquidation while it was being expedited.

Summary and Conclusion

In this research paper, we have developed a comprehensive liquidation methodology for financial assets including semi-liquid structured credit and illiquid fixed assets. We start with a revision of liquidity metrics, developing a Liquidity Value at Risk metric for determining optimal liquidation, then develop a Waterfall approach for an exponential unwind of assets, that we present in a Liquidity Report for a hypothetical Global Rates fund as a final deliverable summary of the techniques developed. We also develop a methodology for a Constrained proportional unwind of assets with heterogeneous liquidity attributes; we encounter a bottleneck paradox in liquidation being dominated by the least liquid asset, even when it is a minority holding. We solve this problem with a Hedged Hibernation approach, whereby we acquire offsetting hedges collateralised and self-financed by the liquidation assets, in order to mitigate the risk of market uncertainty during the liquidation interval. We add this approach to the Liquidity Report to showcase a reduction in overall risk by 80% compared to an outright liquidation.

In delineating the Hedged Hibernation approach to liquidation, we develop a Hedging Toolkit, by borrowing techniques from the Khan Interest Rate Calculus⁵ in order to create Par Yield Durations for both yield and spread for the underlying assets. Next, we derive an optimal hedge ratio for hedge instruments in order to offset the market sensitivities of the outright assets under liquidation, such that we have a market-neutral portfolio with only residual risk in the form of basis risk. For illiquid assets, we further separate the basis risk into an idiosyncratic component and a liquidity premium, and we add an FX Hedge in the form of the Khan Optimal Currency Hedge Ratio⁷.

We review some of the limitations of a Hedged Hibernation Liquidation, establishing main constraints regarding leverage, interim funding, and complete markets regarding alternative hedging instruments. We look at the boundary conditions for a good hedge, in terms of correlation, time to liquidation, underlying volatility of the outright liquidation asset, and the residual basis risk that constitutes the Hedged Liquidity Value at Risk. In summary, a Hedged Hibernation Liquidation with an optimal hedge ratio of:

$$h^* \frac{D_{s_\tau} \sigma_{s_h}}{D_{s_h} \sigma_{s_\tau}} = -w_\pi \rho$$

With a residual basis risk uncertainty: $\sigma_\varepsilon (s - r_T) = w_\pi \, dP_{\%T} = w_\pi \frac{e^{-(r_T+s)T}-1}{(r_T+s)} ds$

Khan Hedged Liquidity Value at Risk LVaR

$$f_t(r_{\pi_T}) = \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{i_c}}{2}}$$

Khan Hedged Liquidity Value at Risk LVaR Constraint for Illiquid Assets

$$y_\tau + \Delta c_{\%} \pi > e^{-r \sqrt{\frac{T_{i_c}}{2}}} \Delta w_{t_0} Z_p \sigma_{\pi_T} \sqrt{\frac{T_{i_c}}{2}}$$

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